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Capítulo 1

Grade 10 - 1st Term

1.1. Grade 10: Discrete Probability Distributions

Complex and non-standard problems solving.

Measure expected monetary value of investments.

- Random variable.
- Probability distributions.
- Cumulative probability distribution.
- Bernoulli random variable.
- Combinations.
- Binomial distribution.
- Negative binomial or Pascal distribution.
- Geometric distribution.
- Mean and variance of discrete distributions.
- Investment portfolio.
- Expected value.

1.1.1. Assessment Criteria

1. Describes random variables considering their characteristics.
2. Interprets simple and cumulative probability distributions.

3. Explains what a Bernoulli random variable is through examples.
4. Identifies a random experiment that follows a binomial distribution.
5. Calculates probabilities where a binomial or negative binomial variable is identified.
6. Establishes cases in which the random variable is geometric.
7. Evaluates probabilities in cases that follow a Poisson distribution.
8. Compares the expected value and variance of the discrete distributions.
9. Deconstructs the concept of an investment portfolio.
10. Measures expected monetary value of investments.

1.1.2. Support Materials

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Capítulo 2

Grade 10 - 2nd Term

2.1. Grade 10: Analysis of Decisions

Decision-making with data relies on payoff tables and criteria that summarise how a decision maker responds to uncertainty. In this topic, we use rules such as Bayes EMV, Maximin and Maximax to compare alternatives across possible states of nature.

- Random variable.
- Probability distributions.
- Subjective probability.
- Mean and variance of discrete distributions.
- Investment portfolio.
- Expected value.
- Define alternatives: list feasible actions.
- Identify states: external outcomes affecting payoffs.
- Assign probabilities: based on data or expert judgment.
- Evaluate: compute expected values or apply rules.
- Bayes EMV: uses probabilities to compute expected value.
- Minimax regret: minimises the largest regret, ignores probabilities.
- Maximax: very optimistic, ignores probabilities.
- Maximin: very cautious, ignores probabilities.

2.1.1. Assessment Criteria

1. Describes the way a problem can be formulated for optimal decision making.
2. Interprets decision alternatives, events, consequences, and states of nature.
3. Builds a payoff table from a description of the problem.
4. Explains the maximax criterion for decision making without probabilities.
5. Summarises the maximin criterion for decision making without probabilities.
6. Develops decision-making strategies using probabilities and the maximum opportunity criterion.
7. Uses probabilities and expected value to analyse a decision-making problem.
8. Compares Bayes, Maximin, and Minimax strategies.
9. Establishes the tree diagram for a decision-making problem.
10. Evaluates all possibilities in a decision-making problem using the strategies presented.

2.1.2. Support Materials

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2.2. Formulas for Analysis of Decisions

2.2.1. Decision-Making Notation

The formulas below use a payoff table with decisions and uncertain future states. Keep the following notation in mind.

- A_j : a decision alternative (strategy) for $j = 1, 2, \dots, n$.
- S_i : a state of nature (future scenario) for $i = 1, 2, \dots, m$.
- $\text{Payoff}(A_j, S_i)$: outcome (profit or loss) if A_j happens and S_i occurs.
- p_i : probability of state S_i (only used when probabilities are given).
- $\max_i$ / $\min_i$: pick the best or worst payoff within a row.
- $\text{Regret}(A_j, S_i)$: missed opportunity compared with the best payoff in state S_i .

2.2.2. Expected Value of a Discrete Random Variable

$$E[X] = \sum_{i=1}^n x_i p_i$$

Where

- X random variable representing uncertain outcomes
- x_i the i -th possible outcome of X
- p_i probability associated with outcome x_i
- n number of possible outcomes

Context Represents the long-run average outcome under repeated identical conditions.

2.2.3. Variance of a Discrete Random Variable

$$\text{Var}(X) = \sum_{i=1}^n (x_i - E[X])^2 p_i$$

Where

- x_i possible outcome
- $E[X]$ expected value of the distribution
- p_i probability of outcome x_i
- n number of outcomes

Context Measures dispersion and quantifies risk or volatility.

2.2.4. Expected Monetary Value (EMV)

$$EMV(A_j) = \sum_{i=1}^m p_i \text{Payoff}(A_j, S_i)$$

Where

- A_j decision alternative (for example a strategy)
- S_i state of nature (future scenario)

- $\text{Payoff}(A_j, S_i)$ monetary result of A_j under state S_i
- p_i probability of state S_i
- m number of states of nature

Context Represents the average monetary performance of a strategy under uncertainty.

2.2.5. Maximin Criterion

$$\text{Maximin}(A_j) = \max_j \left(\min_i \text{Payoff}(A_j, S_i) \right)$$

Where

- A_j decision alternatives
- S_i states of nature
- $\min_i \text{Payoff}(A_j, S_i)$ worst-case payoff of A_j
- outer max selects the highest worst-case result

Context Represents a conservative decision maker seeking safety.

2.2.6. Maximax Criterion

$$\text{Maximax}(A_j) = \max_j \left(\max_i \text{Payoff}(A_j, S_i) \right)$$

Where

- inner max best payoff each alternative can achieve
- outer max selects the alternative with the largest best-case payoff

Context Represents an optimistic decision maker seeking maximum possible gain.

2.2.7. Maximum Opportunity Criterion (Minimax Regret)

$$\text{Regret}(A_j, S_i) = \max_j \text{Payoff}(A_j, S_i) - \text{Payoff}(A_j, S_i)$$

$$\text{Minimax Regret} = \min_j \left(\max_i \text{Regret}(A_j, S_i) \right)$$

Where

- A_j decision alternatives
- S_i states of nature
- $\max_j \text{Payoff}(A_j, S_i)$ best payoff available in state S_i
- $\max_i \text{Regret}(A_j, S_i)$ largest regret for alternative A_j

Context Chooses the alternative that minimizes the worst missed opportunity.

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2.3. Main Examples

2.3.1. Example 1.1

Problem description

A decision maker has three alternatives, A, B and C, and three possible states of nature S_1 , S_2 and S_3 . State probabilities are $P(S_1) = 0,30$, $P(S_2) = 0,40$ and $P(S_3) = 0,30$. The payoffs (in generic units) are:

- Alternative A: $S_1 = 150$, $S_2 = 80$, $S_3 = -20$.
- Alternative B: $S_1 = 110$, $S_2 = 90$, $S_3 = 40$.
- Alternative C: $S_1 = 70$, $S_2 = 70$, $S_3 = 60$.

(C2) Interpreting the parts of the decision problem

We identify the components of the decision situation.

Element	Description
Decision alternatives	Alternatives A, B, C
States of nature	S_1 , S_2 , S_3
Fortuitous events	The state of nature that occurs, unknown at the time of the decision
Consequences	Payoff (generic units) for each alternative–state pair
Probabilities	$P(S_1) = 0,30$, $P(S_2) = 0,40$, $P(S_3) = 0,30$

(C3) Building the payoff table from the description

From the description we build the payoff table.

State of nature	Probability	Alternative A	Alternative B	Alternative C
S_1	0.30	150	110	70
S_2	0.40	80	90	70
S_3	0.30	-20	40	60

(C4) Maximax rule decision making without probabilities

We ignore probabilities and focus on the best payoff for each alternative.

$$\max_i \text{Payoff}(A, S_i) = 150$$

$$\max_i \text{Payoff}(B, S_i) = 110$$

$$\max_i \text{Payoff}(C, S_i) = 70$$

$$\max\{150, 110, 70\} = 150$$

The largest of these is 150, so the Maximax rule recommends Alternative A.

(C5) Maximin rule decision making without probabilities

We focus on the worst payoff for each alternative.

$$\min_i \text{Payoff}(A, S_i) = -20$$

$$\min_i \text{Payoff}(B, S_i) = 40$$

$$\min_i \text{Payoff}(C, S_i) = 60$$

We then select the highest of these worst outcomes:

$$\max\{-20, 40, 60\} = 60$$

The Maximin rule recommends Alternative C.

(C6) Maximum opportunity criterion minimax regret

We compute the regret table by comparing each payoff to the best payoff in each state.

Best payoff in each state

$$S1 : \max\{150, 110, 70\} = 150$$

$$S2 : \max\{80, 90, 70\} = 90$$

$$S3 : \max\{-20, 40, 60\} = 60$$

Regrets are best payoff minus actual payoff.

State of Nature	Alternative A	Alternative B	Alternative C
S_1	$150 - 150 = 0$	$150 - 110 = 40$	$150 - 70 = 80$
S_2	$90 - 80 = 10$	$90 - 90 = 0$	$90 - 70 = 20$
S_3	$60 - (-20) = 80$	$60 - 40 = 20$	$60 - 60 = 0$

Maximum regret for each alternative

$$\max_i \text{Regret}(A, S_i) = \max\{0, 10, 80\} = 80$$

$$\max_i \text{Regret}(B, S_i) = \max\{40, 0, 20\} = 40$$

$$\max_i \text{Regret}(C, S_i) = \max\{80, 20, 0\} = 80$$

$$\min\{80, 40, 80\} = 40$$

The minimax regret rule chooses Alternative B.

(C7) Using probabilities and expected value Bayes decision rule

We compute the expected monetary value (EMV) for each alternative.

For Alternative A

$$\begin{aligned} EMV_A &= 0,30(150) + 0,40(80) + 0,30(-20) \\ &= 45 + 32 - 6 = 71 \end{aligned}$$

For Alternative B

$$\begin{aligned} EMV_B &= 0,30(110) + 0,40(90) + 0,30(40) \\ &= 33 + 36 + 12 = 81 \end{aligned}$$

For Alternative C

$$\begin{aligned} EMV_C &= 0,30(70) + 0,40(70) + 0,30(60) \\ &= 21 + 28 + 18 = 67 \\ \max\{71, 81, 67\} &= 81 \end{aligned}$$

Under the Bayes EMV rule, the highest EMV is 81, so the rule recommends Alternative B.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

Summary of recommendations from the different rules

- Bayes EMV recommends Alternative B.
- Minimax regret recommends Alternative B.
- Maximax recommends Alternative A.
- Maximin recommends Alternative C.

The EMV and minimax regret rules both favor the same alternative, while Maximax and Maximin highlight different extremes of the payoff table.

(C9) Tree diagram of the decision-making problem

We represent the same situation as a decision tree.

This diagram summarises the possible states and payoffs for each alternative.

(C10) Evaluating all possibilities in the decision problem

- Alternative A maximises the best payoff but performs poorly in worst-case and regret terms.
- Alternative C protects against the worst case but yields lower expected value.
- Alternative B is favoured by both expected value and minimax regret.

Overall, the comparison shows how the choice depends on risk attitude, with Alternative B providing a balanced option.

(C1) Formulating the final decision and summarising all criteria

Summary of final recommendations across all decision criteria

- Maximax favors Alternative A, prioritising the highest possible payoff.
- Maximin favors Alternative C, emphasising protection against the worst case.
- Minimax regret favors Alternative B, limiting potential regret.
- Bayes EMV favors Alternative B, maximising expected value.

Considering all criteria together, Alternative B represents the most coherent final decision. It is supported by both expected value and minimax regret and avoids extreme downside outcomes.

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2.3.2. Example 1.2

Problem description

A technology firm is planning its strategy for the next year and must choose exactly one of three possible options. One option is to innovate (Strategy A) by launching a new, cutting-edge product. Another option is to improve the current product line (Strategy B) by upgrading and refining what the firm already sells. The third option is to license an external technology (Strategy C) instead of developing new technology in-house.

The firm knows that future market demand is uncertain and can be in one of three states high demand, moderate demand or low demand. Based on market research, management estimates that the probability of high demand is 0,30, the probability of moderate demand is 0,40 and the probability of low demand is 0,30. These probabilities are treated as the best available information before any decision is made.

For each combination of strategy and demand, the firm has an estimate of the profit measured in thousands of dollars. If demand is high, the expected profits are 150 for Innovate (A), 110 for Improve (B) and 70 for License (C). If demand is moderate, the expected profits are 80 for Innovate (A), 90 for Improve (B) and 70 for License (C). If demand is low, the expected profits are –20 for Innovate (A), 40 for Improve (B) and 60 for License (C). The firm wants to analyse this decision using several decision-making rules, including expected value, maximax, maximin and the maximum opportunity minimax regret criterion, and then determine which strategy provides the most coherent choice given its view of risk and the available probability information.

(C2) Interpreting the parts of the decision problem

We identify the key elements and summarise them in a table.

Element	Description
Decision alternatives	Innovate (A), Improve (B), License (C)
States of nature	High demand, Moderate demand, Low demand
Fortuitous events	The actual market demand that will occur, which is unknown when deciding
Consequences	Profit in thousands of dollars for each strategy–demand combination
Probabilities	High demand 0,30, Moderate demand 0,40, Low demand 0,30

(C3) Building the payoff table from the description

From the problem description, we construct the payoff table (profits in thousands of dollars).

State of Nature	Probability	Innovate (A)	Improve (B)	License (C)
High demand	0,30	150	110	70
Moderate demand	0,40	80	90	70
Low demand	0,30	-20	40	60

(C4) Maximax rule decision making without probabilities

We ignore probabilities and focus only on the best possible payoff for each strategy.

Best payoff for each strategy

$$\max_i \text{Payoff}(A, S_i) = 150k$$

$$\max_i \text{Payoff}(B, S_i) = 110k$$

$$\max_i \text{Payoff}(C, S_i) = 70k$$

The largest of these is $150k$, so the Maximax rule recommends *Innovate (A)*.

(C5) Maximin rule decision making without probabilities

Now we ignore probabilities and focus on the worst-case payoff of each strategy.

Worst payoff for each strategy

$$\min_i \text{Payoff}(A, S_i) = -20k$$

$$\min_i \text{Payoff}(B, S_i) = 40k$$

$$\min_i \text{Payoff}(C, S_i) = 60k$$

We then pick the strategy with the highest of these worst outcomes

$$\max\{-20, 40, 60\} = 60k$$

The Maximin rule recommends *License (C)*.

(C6) Maximum opportunity criterion minimax regret

We construct the regret (opportunity loss) table. For each state of nature, we compare each payoff with the best payoff in that state.

Best payoff in each state

$$\text{High demand : } \max\{150, 110, 70\} = 150k$$

$$\text{Moderate demand : } \max\{80, 90, 70\} = 90k$$

$$\text{Low demand : } \max\{-20, 40, 60\} = 60k$$

Regrets are computed as best payoff minus actual payoff.

State of Nature	Innovate (A)	Improve (B)	License (C)
High demand	$150 - 150 = 0$	$150 - 110 = 40$	$150 - 70 = 80$
Moderate demand	$90 - 80 = 10$	$90 - 90 = 0$	$90 - 70 = 20$
Low demand	$60 - (-20) = 80$	$60 - 40 = 20$	$60 - 60 = 0$

Maximum regret for each strategy

$$\max_i \text{Regret}(A, S_i) = \max\{0, 10, 80\} = 80$$

$$\max_i \text{Regret}(B, S_i) = \max\{40, 0, 20\} = 40$$

$$\max_i \text{Regret}(C, S_i) = \max\{80, 20, 0\} = 80$$

The minimax regret rule chooses the strategy with the smallest maximum regret, so the maximum opportunity criterion recommends *Improve (B)*.

(C7) Using probabilities and expected value Bayes decision rule

We apply the Expected Monetary Value (EMV) rule. For each strategy, we multiply each payoff by the probability of its state and add the results.

For Innovate (A)

$$\begin{aligned}EMV_A &= 0,30(150) + 0,40(80) + 0,30(-20) \\&= 0,30 \cdot 150 + 0,40 \cdot 80 + 0,30 \cdot (-20) \\&= 45 + 32 - 6 = 71k\end{aligned}$$

For Improve (B)

$$\begin{aligned}EMV_B &= 0,30(110) + 0,40(90) + 0,30(40) \\&= 0,30 \cdot 110 + 0,40 \cdot 90 + 0,30 \cdot 40 \\&= 33 + 36 + 12 = 81k\end{aligned}$$

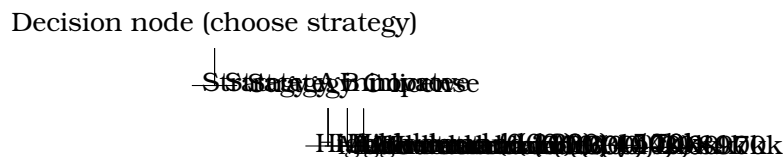
For License (C)

$$\begin{aligned}EMV_C &= 0,30(70) + 0,40(70) + 0,30(60) \\&= 0,30 \cdot 70 + 0,40 \cdot 70 + 0,30 \cdot 60 \\&= 21 + 28 + 18 = 67k\end{aligned}$$

Under the Bayes EMV rule, the best strategy is the one with the highest EMV, so EMV recommends *Improve (B)* with 81k.

(C9) Tree diagram of the decision-making problem

We represent the same situation as a decision tree.



This diagram helps visualise all possible paths decisions followed by demand realisations and the associated payoffs.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

Summary of recommendations from the different rules

- Bayes EMV expected value with probabilities, no experimentation recommends *Improve (B)*.

- Maximum opportunity minimax regret recommends *Improve (B)*.
- Maximax very optimistic, ignores probabilities recommends *Innovate (A)*.
- Maximin very cautious, ignores probabilities recommends *License (C)*.

Bayes and minimax regret use the full payoff structure (and, in the case of Bayes, the probabilities) to favor a more balanced option, whereas Maximax and Maximin focus only on the most extreme possible outcomes.

(C10) Evaluating all possibilities in the decision problem

Considering all the strategies and rules together, the firm can reflect on the consequences of each possible choice. Innovate (A) offers the highest possible payoff but also the lowest worst-case payoff and a high maximum regret. License (C) protects strongly against the worst case but sacrifices expected profit and is not favoured by EMV or regret. Improve (B) is the only strategy that is simultaneously supported by expected value and by the minimax regret rule. This evaluation shows how different attitudes toward risk and regret lead to different recommended actions, and highlights Improve (B) as a compromise between risk and return.

(C1) Formulating the final decision and summarising all criteria

Based on each of the worked decision criteria, the recommended decision would be as follows.

If the firm decides according to the Maximax rule (C4) and is willing to act with a very optimistic, risk-seeking attitude, it should choose *Innovate (A)* because this strategy has the highest possible payoff among all options.

If the firm decides according to the Maximin rule (C5) and adopts a very conservative, worst-case oriented perspective, it should choose *License (C)* since this strategy has the strongest worst-case payoff.

If the firm decides according to the maximum opportunity minimax regret rule (C6) and wants to minimise the largest possible regret, it should choose *Improve (B)* because this strategy has the smallest maximum regret.

If the firm decides according to the Bayes EMV rule (C7) using the estimated probabilities and focusing on long-run average profit, it should again choose *Improve (B)* because it has the highest expected monetary value.

The decision tree (C9) does not change the recommendation but confirms, in a visual and systematic way, the same payoffs and probabilities used in these rules.

Finally, the comparison in (C8) and the global evaluation in (C10) show that each rule corresponds to a particular attitude toward risk and uncertainty.

Taking all criteria together, a coherent summary conclusion is that the strategy *Improve (B)* provides the most balanced decision for this firm. It is favoured by the expected value approach and by the minimax regret criterion, it avoids the very poor worst case of Innovate (A) and it still offers better long-run performance than License (C). Therefore, in the context of this problem and given the available information, the firm should formulate its final decision as adopting the strategy *Improve (B)* for the next year.

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2.3.3. Example 2.1

Problem description

A decision maker chooses among three alternatives A, B and C. The outcome depends on two independent conditions. The value-related condition has two levels V_1 and V_2 with probabilities 0.30 and 0.70. The cost-related condition has two levels C_1 and C_2 with probabilities 0.60 and 0.40. For each alternative, there is a value component associated with the value condition and a cost component associated with the cost condition. Payoff is defined as value minus cost.

The value component for alternatives A, B and C is 7000, 5000 and 4800 under V_1 , and 3200, 3600 and 3900 under V_2 . The cost component for alternatives A, B and C is 2300, 1600 and 2000 under C_1 , and 3500, 2600 and 2700 under C_2 . Because the conditions are independent, the combined states of nature are S_1 – S_4 with probabilities 0.18, 0.12, 0.42 and 0.28. Payoffs for each combined state are shown in the payoff table below.

(C2) Interpreting the parts of the decision problem

We interpret the narrative and identify the key components.

Element	Description
Decision alternatives	Alternative A, Alternative B, Alternative C
States of nature	Combined condition scenarios S_1 – S_4
Fortuitous events	The actual condition levels that occur
Consequences	Payoff for each alternative and combined scenario
Probabilities	Value condition: 0.30 V_1 , 0.70 V_2 ; Cost condition: 0.60 C_1 , 0.40 C_2 ; independent events

(C3) Building the payoff table from revenue and expense information

From the problem description, we compute the combined-state probabilities and build the payoff table.

We define four combined states of nature: S_1 is V_1 with C_1 , S_2 is V_1 with C_2 , S_3 is V_2 with C_1 and S_4 is V_2 with C_2 . The joint probabilities are

$$P(S_1) = 0,30 \cdot 0,60 = 0,18,$$

$$P(S_2) = 0,30 \cdot 0,40 = 0,12,$$

$$P(S_3) = 0,70 \cdot 0,60 = 0,42,$$

$$P(S_4) = 0,70 \cdot 0,40 = 0,28.$$

State of Nature	Probability	Alternative A	Alternative B	Alternative C
$S_1 (V_1, C_1)$	0.18	$7000 - 2300 = 4700$	$5000 - 1600 = 3400$	$4800 - 2000 = 2800$
$S_2 (V_1, C_2)$	0.12	$7000 - 3500 = 3500$	$5000 - 2600 = 2400$	$4800 - 2700 = 2100$
$S_3 (V_2, C_1)$	0.42	$3200 - 2300 = 900$	$3600 - 1600 = 2000$	$3900 - 2000 = 1900$
$S_4 (V_2, C_2)$	0.28	$3200 - 3500 = -300$	$3600 - 2600 = 1000$	$3900 - 2700 = 1200$

(C4) Maximax rule for decision making without probabilities

We ignore probabilities and identify the best payoff each alternative can achieve.

Best payoff for each alternative

$$\max_i \text{Payoff}(A, S_i) = 4700$$

$$\max_i \text{Payoff}(B, S_i) = 3400$$

$$\max_i \text{Payoff}(C, S_i) = 2800$$

$$\max\{4700, 3400, 2800\} = 4700$$

The Maximax rule recommends Alternative A.

(C5) Maximin rule for decision making without probabilities

We focus on the worst payoff for each alternative.

Worst payoff for each alternative

$$\min_i \text{Payoff}(A, S_i) = -300$$

$$\min_i \text{Payoff}(B, S_i) = 1000$$

$$\min_i \text{Payoff}(C, S_i) = 1200$$

We then select the highest of these worst outcomes:

$$\max\{-300, 1000, 1200\} = 1200$$

The Maximin rule recommends Alternative C.

(C6) Maximum opportunity criterion (minimax regret)

We compute regrets by comparing each payoff to the best payoff in each state.

Best payoff in each state

$$S_1 : \max\{4700, 3400, 2800\} = 4700$$

$$S_2 : \max\{3500, 2400, 2100\} = 3500$$

$$S_3 : \max\{900, 2000, 1900\} = 2000$$

$$S_4 : \max\{-300, 1000, 1200\} = 1200$$

Regrets are best payoff minus actual payoff.

State of Nature	Alternative A	Alternative B	Alternative C
S_1	$4700 - 4700 = 0$	$4700 - 3400 = 1300$	$4700 - 2800 = 1900$
S_2	$3500 - 3500 = 0$	$3500 - 2400 = 1100$	$3500 - 2100 = 1400$
S_3	$2000 - 900 = 1100$	$2000 - 2000 = 0$	$2000 - 1900 = 100$
S_4	$1200 - (-300) = 1500$	$1200 - 1000 = 200$	$1200 - 1200 = 0$

Maximum regret for each alternative

$$\max_i \text{Regret}(A, S_i) = \max\{0, 0, 1100, 1500\} = 1500$$

$$\max_i \text{Regret}(B, S_i) = \max\{1300, 1100, 0, 200\} = 1300$$

$$\max_i \text{Regret}(C, S_i) = \max\{1900, 1400, 100, 0\} = 1900$$

$$\min\{1500, 1300, 1900\} = 1300$$

The minimax regret rule recommends Alternative B.

(C7) Using probabilities and expected value (Bayes decision rule)

We compute EMV for each alternative by multiplying each payoff by its probability and summing.

For Alternative A

$$\begin{aligned}EMV_A &= 0,18(4700) + 0,12(3500) + 0,42(900) + 0,28(-300) \\ &= 846 + 420 + 378 - 84 = 1560\end{aligned}$$

For Alternative B

$$\begin{aligned}EMV_B &= 0,18(3400) + 0,12(2400) + 0,42(2000) + 0,28(1000) \\ &= 612 + 288 + 840 + 280 = 2020\end{aligned}$$

For Alternative C

$$\begin{aligned}EMV_C &= 0,18(2800) + 0,12(2100) + 0,42(1900) + 0,28(1200) \\ &= 504 + 252 + 798 + 336 = 1890\end{aligned}$$

$$\max\{1560, 2020, 1890\} = 2020$$

Bayes EMV recommends Alternative B.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

Summary of recommendations

- Bayes EMV recommends Alternative B.
- Minimax regret recommends Alternative B.
- Maximax recommends Alternative A.
- Maximin recommends Alternative C.

The rules emphasise different aspects of performance: expected value and regret align on B, while Maximax favors the best possible single outcome (A) and Maximin favors safety (C).

(C9) Tree diagram of the decision-making problem

We represent the combined-state problem as a decision tree.

This diagram summarises the possible combined states and payoffs for each alternative.

(C10) Evaluating all possibilities in the decision problem

- Alternative A is attractive for its best-case payoff but has the poorest worst-case outcome.
- Alternative C protects the worst case but has lower expected value than B.
- Alternative B is supported by both expected value and minimax regret.

Overall, Alternative B offers a balanced decision when probabilities are available.

(C1) Formulating the final decision and summarising all criteria

Summary across all criteria

- Maximax favors Alternative A, prioritising the highest possible payoff.
- Maximin favors Alternative C, emphasising protection against the worst case.
- Minimax regret favors Alternative B, limiting potential regret.
- Bayes EMV favors Alternative B, maximising expected value.

Considering all criteria together, Alternative B represents the most coherent final decision because it performs well on expected value and regret and avoids extreme downside risk.

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2.3.4. Example 2.2

Problem description

A Colombian agro-export firm that currently sells fresh avocados and coffee in several markets is planning a long-term investment strategy. Management is considering three mutually exclusive options for the next five years. The first possibility is to build a modern avocado processing plant focused on the European Union market, which will be referred to as Strategy A and described as the EU avocado plant. The second possibility is to concentrate on raw coffee exports to the United States, improving logistics and marketing for that specific product, which will be referred to as Strategy B or US coffee focus. The third possibility is to diversify its exports across several regional markets in Latin America and to maintain a smaller presence in both the European Union and the United States, which will be referred to as Strategy C or regional diversification.

The firm is worried about uncertainty in two independent aspects that affect its profit. The first aspect is global demand for its export products, which affects revenue. The second

aspect is production and logistics costs, which affect expenses. Management models global demand with three possible demand scenarios. When world demand is high, the EU avocado plant is expected to generate annual revenue of 7000, the US coffee focus 5000 and regional diversification 4800. When demand is moderate, the EU avocado plant is expected to generate revenue of 4800, the US coffee focus 4200 and regional diversification 4300. When demand is weak, the EU avocado plant is expected to generate revenue of 3200, the US coffee focus 3600 and regional diversification 3900. These revenue figures are measured in thousands of US dollars per year. The firm estimates that the probability of high demand is 0.30, the probability of moderate demand is 0.45 and the probability of weak demand is 0.25.

Independent of demand, the firm also considers two possible cost scenarios for production and logistics. In a favourable cost scenario with low input and shipping costs, yearly expenses are expected to be 2300 for the EU avocado plant, 1600 for the US coffee focus and 2000 for regional diversification. In an unfavourable cost scenario with high fuel prices and shipping costs, yearly expenses are expected to be 3500 for the EU avocado plant, 2600 for the US coffee focus and 2700 for regional diversification. These expense figures are also measured in thousands of US dollars per year. The firm estimates that the probability of favourable costs is 0.60 and the probability of unfavourable costs is 0.40, and it assumes that demand conditions and cost conditions are independent.

Profit in each year is defined as revenue minus expenses. Therefore, profit for each strategy in any given year will be determined by the combination of the demand scenario that occurs and the cost scenario that occurs. Because the firm considers the demand event and the cost event to be independent, the probability of any combined scenario is the product of the probability of the corresponding demand level and the probability of the corresponding cost level. Management wants to analyse these three strategies using different decision rules, including Maximax, Maximin, the maximum opportunity minimax regret criterion and the Bayes expected value rule, and then decide which strategy is most coherent given its views about risk and its probability assessments for demand and costs.

(C2) Interpreting the parts of the decision problem

We interpret the narrative and identify the key components of the decision situation.

(C3) Building the payoff table from revenue and expense information

From the problem description, we first organise the revenue and expense information for each separate event. All figures are in thousands of US dollars per year.

Element	Description
Decision alternatives	Strategy A is the EU avocado plant, which focuses on processed avocados for the European Union. Strategy B is the US coffee focus, which concentrates on raw coffee exports to the United States. Strategy C is regional diversification, which mixes exports across Latin America and maintains a smaller EU and US presence.
Independent events related to uncertainty	Global demand conditions, which determine revenue, with three levels high, moderate and weak. Cost conditions, which determine expenses, with two levels favourable costs and unfavourable costs.
States of nature	Each state of nature is a combination of one demand level and one cost level, for example high demand with favourable costs, moderate demand with unfavourable costs and so on.
Fortuitous events	The actual demand level and the actual cost level that occur in a given year. These outcomes are outside the firm's control and are unknown when the strategy is chosen.
Consequences	Annual profit measured in thousands of US dollars for each strategy, obtained by subtracting the expense level that corresponds to the cost scenario from the revenue level that corresponds to the demand scenario.
Probabilities	Demand probabilities are 0.30 for high demand, 0.45 for moderate demand and 0.25 for weak demand. Cost probabilities are 0.60 for favourable costs and 0.40 for unfavourable costs. The events demand level and cost level are assumed independent, so the probability of any combined state of nature is the product of the corresponding demand probability and cost probability.

Revenue as a function of demand for each strategy:

Expenses as a function of cost conditions for each strategy:

Because demand and cost conditions are independent, a state of nature is described by a pair of events, for example high demand with favourable costs. The probability of each

Demand level	Probability	Strategy A Revenue	Strategy B Revenue	Strategy C Revenue
High demand	0.30	7000	5000	4800
Moderate demand	0.45	4800	4200	4300
Weak demand	0.25	3200	3600	3900

Cost level	Probability	Strategy A Expenses	Strategy B Expenses	Strategy C Expenses
Favourable costs	0.60	2300	1600	2000
Unfavourable costs	0.40	3500	2600	2700

combined state is the product of the probabilities of the corresponding demand and cost levels. Profit is revenue minus expenses for the chosen strategy in that combined state.

We define six combined states of nature: S1 is high demand with favourable costs, S2 is high demand with unfavourable costs, S3 is moderate demand with favourable costs, S4 is moderate demand with unfavourable costs, S5 is weak demand with favourable costs and S6 is weak demand with unfavourable costs. The joint probabilities are

$$P(S1) = 0,30 \cdot 0,60 = 0,18, \quad P(S2) = 0,30 \cdot 0,40 = 0,12, \quad P(S3) = 0,45 \cdot 0,60 = 0,27,$$

$$P(S4) = 0,45 \cdot 0,40 = 0,18, \quad P(S5) = 0,25 \cdot 0,60 = 0,15, \quad P(S6) = 0,25 \cdot 0,40 = 0,10.$$

Profit for each strategy and state of nature is revenue minus expenses. For example, under S3, which is moderate demand and favourable costs, the EU avocado plant has revenue 4800 and expenses 2300, so profit is $4800 - 2300 = 2500$. Under S6, which is weak demand and unfavourable costs, the US coffee focus has revenue 3600 and expenses 2600, so profit is $3600 - 2600 = 1000$.

The payoff table of profit in thousands of US dollars for each combined state is:

(C4) Maximax rule for decision making without probabilities

Under the Maximax rule, the decision maker focuses on the best possible payoff of each strategy and ignores both probabilities and worst cases. We identify, for each strategy, the highest profit among the six states.

$$\max\{4700, 3500, 2500, 1300, 900, -300\} = 4700.$$

State of nature	Combined scenario	Probability	Strategy A Profit	Strategy B Profit	Strategy C Profit
S1	High demand, favourable costs	0.18	4700	3400	2800
S2	High demand, unfavourable costs	0.12	3500	2400	2100
S3	Moderate demand, favourable costs	0.27	2500	2600	2300
S4	Moderate demand, unfavourable costs	0.18	1300	1600	1600
S5	Weak demand, favourable costs	0.15	900	2000	1900
S6	Weak demand, unfavourable costs	0.10	-300	1000	1200

$$\max\{3400, 2400, 2600, 1600, 2000, 1000\} = 3400.$$

$$\max\{2800, 2100, 2300, 1600, 1900, 1200\} = 2800.$$

The largest of these best profits is 4700, which corresponds to strategy A in state S1. According to the Maximax rule, the firm should choose *Strategy A (EU avocado plant)*.

(C5) Maximin rule for decision making without probabilities

Under the Maximin rule, the decision maker looks at the worst profit of each strategy and then chooses the strategy whose worst-case outcome is the strongest.

$$\min\{4700, 3500, 2500, 1300, 900, -300\} = -300.$$

$$\min\{3400, 2400, 2600, 1600, 2000, 1000\} = 1000.$$

$$\min\{2800, 2100, 2300, 1600, 1900, 1200\} = 1200.$$

We compare these worst-case values and choose the highest one,

$$\max\{-300, 1000, 1200\} = 1200.$$

Under the Maximin rule, the firm should choose *Strategy C (regional diversification)*, because it offers the strongest worst-case profit.

(C6) Maximum opportunity criterion (minimax regret)

The maximum opportunity or minimax regret criterion focuses on regret, defined as the loss from not having chosen the best strategy in a given state of nature. For each state, we identify the best profit and then compute regret as the best profit in that state minus the profit of the chosen strategy.

$$S1 \text{ (high demand, favourable costs)} : \max\{4700, 3400, 2800\} = 4700,$$

$$S2 \text{ (high demand, unfavourable costs)} : \max\{3500, 2400, 2100\} = 3500,$$

$$S3 \text{ (moderate demand, favourable costs)} : \max\{2500, 2600, 2300\} = 2600,$$

$$S4 \text{ (moderate demand, unfavourable costs)} : \max\{1300, 1600, 1600\} = 1600,$$

$$S5 \text{ (weak demand, favourable costs)} : \max\{900, 2000, 1900\} = 2000,$$

$$S6 \text{ (weak demand, unfavourable costs)} : \max\{-300, 1000, 1200\} = 1200.$$

The regret table in thousands of US dollars is:

For each strategy, we identify its maximum regret across all states:

$$\max \text{ Regret for A} = \max\{0, 0, 100, 300, 1100, 1500\} = 1500,$$

$$\max \text{ Regret for B} = \max\{1300, 1100, 0, 0, 0, 200\} = 1300,$$

$$\max \text{ Regret for C} = \max\{1900, 1400, 300, 0, 100, 0\} = 1900.$$

The minimax regret rule chooses the strategy with the smallest of these maximum regrets, which is strategy B. The maximum opportunity criterion therefore recommends *Strategy B (US coffee focus)*.

State	Scenario	Strategy A Regret	Strategy B Regret	Strategy C Regret
S1	High demand, favourable costs	$4700 - 4700 = 0$	$4700 - 3400 = 1300$	$4700 - 2800 = 1900$
S2	High demand, unfavourable costs	$3500 - 3500 = 0$	$3500 - 2400 = 1100$	$3500 - 2100 = 1400$
S3	Moderate demand, favourable costs	$2600 - 2500 = 100$	$2600 - 2600 = 0$	$2600 - 2300 = 300$
S4	Moderate demand, unfavourable costs	$1600 - 1300 = 300$	$1600 - 1600 = 0$	$1600 - 1600 = 0$
S5	Weak demand, favourable costs	$2000 - 900 = 1100$	$2000 - 2000 = 0$	$2000 - 1900 = 100$
S6	Weak demand, unfavourable costs	$1200 - (-300) = 1500$	$1200 - 1000 = 200$	$1200 - 1200 = 0$

(C7) Using probabilities and expected value (Bayes decision rule)

The Bayes decision rule with expected monetary value (EMV) multiplies each profit by the probability of its state and adds the results. We use the joint probabilities of S1 to S6.

$$\begin{aligned}
 EMV_A &= 0,18(4700) + 0,12(3500) + 0,27(2500) + 0,18(1300) + 0,15(900) + 0,10(-300) \\
 &= 846 + 420 + 675 + 234 + 135 - 30 = 2280.
 \end{aligned}$$

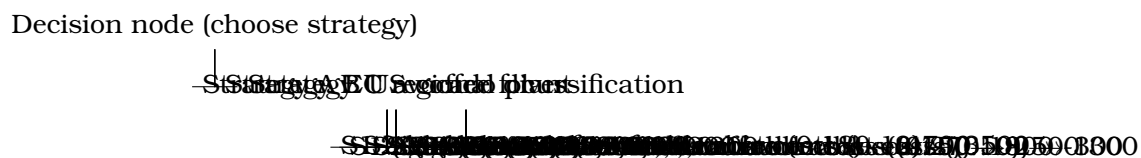
$$\begin{aligned}
 EMV_B &= 0,18(3400) + 0,12(2400) + 0,27(2600) + 0,18(1600) + 0,15(2000) + 0,10(1000) \\
 &= 612 + 288 + 702 + 288 + 300 + 100 = 2290.
 \end{aligned}$$

$$\begin{aligned}
 EMV_C &= 0,18(2800) + 0,12(2100) + 0,27(2300) + 0,18(1600) + 0,15(1900) + 0,10(1200) \\
 &= 504 + 252 + 621 + 288 + 285 + 120 = 2070.
 \end{aligned}$$

These values are expected annual profits in thousands of US dollars. Since $EMV_B = 2290$ is slightly higher than $EMV_A = 2280$ and larger than $EMV_C = 2070$, the Bayes EMV rule recommends *Strategy B (US coffee focus)*.

(C9) Tree diagram of the decision-making problem

The same decision situation can be represented as a decision tree. The firm first chooses a strategy, then the demand and cost outcomes are realised. Because demand and costs are independent, their probabilities multiply along each path. For simplicity, we show the combined states with their joint probabilities and profits.



This diagram makes explicit the sequence of decisions and uncertain events, as well as the joint probabilities and profits associated with each possible path.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

We summarise the recommendations from the different decision rules:

- Maximax, which is very optimistic and focuses on the best possible profit, recommends Strategy A because it can reach 4700.
- Maximin, which is very cautious and focuses on the worst-case profit, recommends Strategy C because its worst case is 1200.
- The maximum opportunity or minimax regret rule, which focuses on limiting the largest possible regret, recommends Strategy B because its maximum regret is 1300, smaller than the maximum regrets of A and C.
- The Bayes EMV rule, which uses the full joint distribution of probabilities and profits, recommends Strategy B because it has the highest expected monetary value.

This comparison shows that rules ignoring probabilities and focusing on extreme outcomes can produce different recommendations from rules that use the full information about the distribution of profits. Strategy A is favoured by a very optimistic view, Strategy C by a very cautious view, and Strategy B by both minimax regret and expected value.

(C10) Evaluating all possibilities in the decision problem

Looking across all rules, the firm can evaluate each strategy in terms of risk and reward. Strategy A, the EU avocado plant, offers the highest possible profit if demand is high and costs are favourable but exposes the firm to a negative profit of -300 when demand is weak

and costs are unfavourable. It also has a relatively large maximum regret, because in some unfavourable states another strategy would have generated much better results. Strategy C, regional diversification, never performs extremely badly or extremely well. It offers the strongest worst-case profit but does not maximise expected profit or minimise regret. Strategy B, the US coffee focus, does not reach the highest possible profit of Strategy A but avoids losses, maintains good results in all combined states and is simultaneously favoured by expected value and by the minimax regret criterion. This evaluation highlights how attitudes towards risk, regret and the use of probabilities lead to different recommended actions, and why Strategy B appears as a compromise between the more aggressive and the more conservative options.

(C1) Formulating the final decision and summarising all criteria

Based on each of the decision criteria used in the solution, the recommended strategy would be as follows. If the firm decides according to the Maximax rule and adopts a highly optimistic and risk-seeking attitude, it should choose Strategy A, the EU avocado plant, because this option can reach the highest profit among all combined states. If the firm decides according to the Maximin rule and wants to protect itself against the worst possible profit, it should choose Strategy C, regional diversification, because this option has the strongest worst-case outcome. If the firm uses the maximum opportunity or minimax regret criterion and cares mainly about limiting the largest possible regret from a wrong choice, it should choose Strategy B, the US coffee focus, because this option has the smallest maximum regret. If the firm trusts the estimated probabilities for demand and cost conditions and uses the Bayes expected value rule to maximise expected annual profit, it should again choose Strategy B, since it has the highest expected monetary value.

The decision tree representation confirms the structure of the problem and helps visualise the role of the two independent events demand and costs, while the comparison and the overall evaluation clarify how each rule reflects a particular attitude to risk and uncertainty. Taking all criteria together, a coherent summary conclusion is that Strategy B, the US coffee focus, is the most balanced choice for this firm. It avoids the potential losses of Strategy A in adverse states, performs better than Strategy C in terms of expected profit and at the same time keeps the maximum regret at a relatively low level. For a firm that wants to use probability information responsibly and avoid extreme risk while still pursuing strong expected returns, formulating the final decision as choosing Strategy B is consistent with the overall analysis of the decision-making problem.

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2.3.5. Example 2.3

Problem description

A Colombian agro-export firm that sells fruit-based products and coffee is planning its strategy for the next five years. Management wants to separate the way it makes decisions about revenue from the way it makes decisions about expenses. For the revenue side, the firm is considering two different market and pricing strategies. The first revenue strategy is a premium positioning aimed at high-income consumers and European specialty stores, with higher prices but a stronger dependence on favourable demand conditions. This will be called the premium revenue strategy. The second revenue strategy is a volume-based strategy focused on larger retail chains and regional markets, with more moderate prices but more stable volumes. This will be called the volume revenue strategy.

For the expense side, the firm must choose how to organise its production and logistics. One option is to invest in a more automated and contracted logistics system, which requires higher fixed costs but reduces exposure to large increases in variable costs. This will be called the stable cost strategy. The other option is to rely on flexible outsourcing and spot contracts for transport and inputs, which can be cheaper when global conditions are calm but much more expensive when there are cost shocks. This will be called the flexible cost strategy. The firm must choose one revenue strategy and one cost strategy, so it effectively chooses a combination such as premium revenue with stable costs or volume revenue with flexible costs.

The firm faces two independent sources of uncertainty. The first is global demand for its products, which can be high, moderate or weak. When demand is high, if the firm chooses the premium revenue strategy it expects annual revenue of 6000, while with the volume revenue strategy it expects annual revenue of 5500. When demand is moderate, the premium revenue strategy is expected to generate revenue of 4500 and the volume revenue strategy revenue of 4300. When demand is weak, the premium revenue strategy is expected to generate revenue of 3200 and the volume revenue strategy revenue of 3800. All revenues are measured in thousands of US dollars per year. Based on market analysis, management estimates that the probability of high demand is 0.30, the probability of moderate demand is 0.45 and the probability of weak demand is 0.25.

The second source of uncertainty is the global cost environment, which affects production and logistics expenses. The firm considers two possible cost conditions. In a favourable cost environment, with relatively low input and shipping costs, yearly expenses are expected to be 2000 if the firm chooses the stable cost strategy and 1500 if it chooses the flexible cost strategy. In an unfavourable cost environment, with high fuel prices and congested shipping routes, yearly expenses are expected to be 2600 under the stable cost strategy

and 3400 under the flexible cost strategy. These expense values are also measured in thousands of US dollars per year. The firm estimates that the probability of favourable costs is 0.60 and the probability of unfavourable costs is 0.40. It assumes that global demand conditions and the global cost environment are independent.

Annual profit is defined as revenue minus expenses. Therefore, for each year, profit is determined by the combination of the demand scenario that occurs, the cost scenario that occurs and the pair of decisions chosen by the firm (one decision for revenue and another for costs). Because demand and cost conditions are independent, the probability of any combined state of nature is the product of the probability of the corresponding demand level and the probability of the corresponding cost level. Management wants to compare the different revenue–cost combinations using several decision rules, including Maximax, Maximin, the maximum opportunity (minimax regret) criterion and the Bayes expected value rule, and then decide which combination of revenue strategy and cost strategy is most coherent given its preferences about risk and its probability assessments.

(C2) Interpreting the parts of the decision problem

We interpret the narrative and identify the key components of the decision situation.

(C3) Building the payoff table from the revenue and expense information

From the problem description, we first organise the revenue and expense information for each separate event and decision. All amounts are in thousands of US dollars per year.

Revenue as a function of demand for each revenue strategy:

Expenses as a function of cost conditions for each cost strategy:

Because demand and cost conditions are independent, a state of nature is described by a pair of events. We define six combined states: S1 is high demand with favourable costs, S2 is high demand with unfavourable costs, S3 is moderate demand with favourable costs, S4 is moderate demand with unfavourable costs, S5 is weak demand with favourable costs and S6 is weak demand with unfavourable costs. The joint probabilities are

$$P(S1) = 0,30 \cdot 0,60 = 0,18,$$

$$P(S2) = 0,30 \cdot 0,40 = 0,12,$$

$$P(S3) = 0,45 \cdot 0,60 = 0,27,$$

$$P(S4) = 0,45 \cdot 0,40 = 0,18,$$

$$P(S5) = 0,25 \cdot 0,60 = 0,15,$$

$$P(S6) = 0,25 \cdot 0,40 = 0,10.$$

Element	Description
Revenue decision alternatives	Premium revenue strategy (higher prices, more sensitive to demand). Volume revenue strategy (moderate prices, more stable volumes).
Cost decision alternatives	Stable cost strategy (automation and long-term contracts, more stable expenses). Flexible cost strategy (outsourcing and spot contracts, cheaper in calm periods but risky under cost shocks).
Combined decision alternatives	A: Premium revenue + stable costs. B: Premium revenue + flexible costs. C: Volume revenue + stable costs. D: Volume revenue + flexible costs.
Independent events	Demand event with three levels high, moderate and weak, which determines revenue. Cost event with two levels favourable and unfavourable, which determines expenses.
States of nature	Each state of nature is a combination of one demand level and one cost level, for example high demand with favourable costs or weak demand with unfavourable costs.
Probabilities	Demand probabilities are 0.30 for high demand, 0.45 for moderate demand and 0.25 for weak demand. Cost probabilities are 0.60 for favourable costs and 0.40 for unfavourable costs. Demand and cost events are independent, so each combined state's probability is the product of the relevant demand probability and cost probability.
Consequences	Annual profit in thousands of US dollars, calculated as revenue minus expenses, for each combined revenue–cost decision and each combined state of nature.

Profit for each combined decision and state of nature is revenue minus expenses. We label the combined decisions as follows: A is premium revenue with stable costs, B is premium revenue with flexible costs, C is volume revenue with stable costs and D is volume revenue with flexible costs. For example, in S1 there is high demand and favourable costs. Under decision A, revenue is 6000 and expenses are 2000, so profit is $6000 - 2000 = 4000$. Under

Demand level	Probability	Premium revenue strategy	Volume revenue strategy
High	0.30	6000	5500
Moderate	0.45	4500	4300
Weak	0.25	3200	3800

Cost level	Probability	Stable cost strategy	Flexible cost strategy
Favourable	0.60	2000	1500
Unfavourable	0.40	2600	3400

decision B, revenue is 6000 and expenses are 1500, so profit is $6000 - 1500 = 4500$. In S6 there is weak demand and unfavourable costs. Under decision B, revenue is 3200 and expenses are 3400, so profit is $3200 - 3400 = -200$.

The resulting payoff table of profit (in thousands of US dollars) is:

(C4) Maximax rule for decision making without probabilities

Under the Maximax rule, the decision maker focuses on the best possible profit of each combined decision and ignores probabilities and worst cases.

$$\max\{4000, 3400, 2500, 1900, 1200, 600\} = 4000.$$

$$\max\{4500, 2600, 3000, 1100, 1700, -200\} = 4500.$$

$$\max\{3500, 2900, 2300, 1700, 1800, 1200\} = 3500.$$

$$\max\{4000, 2100, 2800, 900, 2300, 400\} = 4000.$$

The largest of these best profits is 4500, which corresponds to decision B in state S1. According to the Maximax rule, the firm should choose *decision B (premium revenue with flexible costs)*.

(C5) Maximin rule for decision making without probabilities

Under the Maximin rule, the decision maker looks at the worst profit of each combined decision and then chooses the decision whose worst-case outcome is the strongest.

$$\min\{4000, 3400, 2500, 1900, 1200, 600\} = 600.$$

State	Combined scenario	Probability	A: Premium stable	B: Premium flexible	C: Volume stable	D: Volume flexible
S1	High demand, favourable costs	0.18	4000	4500	3500	4000
S2	High demand, unfavourable costs	0.12	3400	2600	2900	2100
S3	Moderate demand, favourable costs	0.27	2500	3000	2300	2800
S4	Moderate demand, unfavourable costs	0.18	1900	1100	1700	900
S5	Weak demand, favourable costs	0.15	1200	1700	1800	2300
S6	Weak demand, unfavourable costs	0.10	600	-200	1200	400

$$\min\{4500, 2600, 3000, 1100, 1700, -200\} = -200.$$

$$\min\{3500, 2900, 2300, 1700, 1800, 1200\} = 1200.$$

$$\min\{4000, 2100, 2800, 900, 2300, 400\} = 400.$$

We compare these worst-case values and choose the highest one,

$$\max\{600, -200, 1200, 400\} = 1200.$$

Under the Maximin rule, the firm should choose *decision C (volume revenue with stable costs)*, because it offers the strongest worst-case profit.

(C6) Maximum opportunity criterion (minimax regret)

The maximum opportunity or minimax regret criterion focuses on regret, defined as the loss from not having chosen the best decision in a given state of nature. For each state, we identify the best profit and then compute regret as the best profit in that state minus the profit of the chosen decision.

$$S1: \max\{4000, 4500, 3500, 4000\} = 4500,$$

$$S2: \max\{3400, 2600, 2900, 2100\} = 3400,$$

$$S3: \max\{2500, 3000, 2300, 2800\} = 3000,$$

$$S4: \max\{1900, 1100, 1700, 900\} = 1900,$$

$$S5: \max\{1200, 1700, 1800, 2300\} = 2300,$$

$$S6: \max\{600, -200, 1200, 400\} = 1200.$$

The regret table in thousands of US dollars is:

For each decision, we identify its maximum regret across all states:

$$\max \text{Regret}(A) = \max\{500, 0, 500, 0, 1100, 600\} = 1100,$$

$$\max \text{Regret}(B) = \max\{0, 800, 0, 800, 600, 1400\} = 1400,$$

$$\max \text{Regret}(C) = \max\{1000, 500, 700, 200, 500, 0\} = 1000,$$

$$\max \text{Regret}(D) = \max\{500, 1300, 200, 1000, 0, 800\} = 1300.$$

The minimax regret rule chooses the decision with the smallest of these maximum regrets, which is decision C. The maximum opportunity criterion therefore recommends *decision C (volume revenue with stable costs)*.

(C7) Using probabilities and expected value (Bayes decision rule)

The Bayes decision rule with expected monetary value (EMV) multiplies each profit by the probability of its state and adds the results. We use the joint probabilities of S1 to S6.

$$\begin{aligned} EMV_A &= 0,18(4000) + 0,12(3400) + 0,27(2500) + 0,18(1900) + 0,15(1200) + 0,10(600) \\ &= 720 + 408 + 675 + 342 + 180 + 60 = 2385. \end{aligned}$$

$$\begin{aligned} EMV_B &= 0,18(4500) + 0,12(2600) + 0,27(3000) + 0,18(1100) + 0,15(1700) + 0,10(-200) \\ &= 810 + 312 + 810 + 198 + 255 - 20 = 2365. \end{aligned}$$

$$\begin{aligned} EMV_C &= 0,18(3500) + 0,12(2900) + 0,27(2300) + 0,18(1700) + 0,15(1800) + 0,10(1200) \\ &= 630 + 348 + 621 + 306 + 270 + 120 = 2295. \end{aligned}$$

State	Scenario	A Regret	B Regret	C Regret	D Regret
S1	High demand, favourable costs	$4500 - 4000 = 500$	$4500 - 4500 = 0$	$4500 - 3500 = 1000$	$4500 - 4000 = 500$
S2	High demand, unfavourable costs	$3400 - 3400 = 0$	$3400 - 2600 = 800$	$3400 - 2900 = 500$	$3400 - 2100 = 1300$
S3	Moderate demand, favourable costs	$3000 - 2500 = 500$	$3000 - 3000 = 0$	$3000 - 2300 = 700$	$3000 - 2800 = 200$
S4	Moderate demand, unfavourable costs	$1900 - 1900 = 0$	$1900 - 1100 = 800$	$1900 - 1700 = 200$	$1900 - 900 = 1000$
S5	Weak demand, favourable costs	$2300 - 1200 = 1100$	$2300 - 1700 = 600$	$2300 - 1800 = 500$	$2300 - 2300 = 0$
S6	Weak demand, unfavourable costs	$1200 - 600 = 600$	$1200 - (-200) = 1400$	$1200 - 1200 = 0$	$1200 - 400 = 800$

$$\begin{aligned}
 EMV_D &= 0,18(4000) + 0,12(2100) + 0,27(2800) + 0,18(900) + 0,15(2300) + 0,10(400) \\
 &= 720 + 252 + 756 + 162 + 345 + 40 = 2275.
 \end{aligned}$$

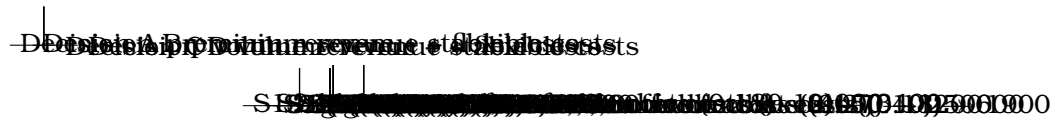
These values are expected annual profits in thousands of US dollars. Since $EMV_A = 2385$ is the highest expected value, the Bayes EMV rule recommends *decision A (premium revenue with stable costs)*.

(C9) Tree diagram of the decision-making problem

The same decision situation can be represented as a decision tree. The firm first chooses a combined decision (revenue strategy and cost strategy), then the demand level is realised and then the cost level is realised. Because demand and costs are independent, their

probabilities multiply along each path. For simplicity, we show the combined states with their joint probabilities and profits.

Decision node (choose combined decision)



This diagram makes explicit the sequence of decisions and uncertain events, as well as the joint probabilities and profits associated with each possible path.

(C8) Comparing Bayes, Maximin, Maximax and minimax regret

We summarise the recommendations from the different decision rules:

- Maximax, which is very optimistic and focuses on the best possible profit, recommends decision B (premium revenue + flexible costs) because it can reach 4500.
- Maximin, which is very cautious and focuses on the worst-case profit, recommends decision C (volume revenue + stable costs) because its worst case is 1200.
- The maximum opportunity or minimax regret rule, which focuses on limiting the largest possible regret, recommends decision C because its maximum regret is 1000, smaller than the maximum regrets of A, B and D.
- The Bayes EMV rule, which uses the full joint distribution of probabilities and profits, recommends decision A (premium revenue + stable costs) because it has the highest expected monetary value, 2385.

The comparison shows that rules ignoring probabilities and focusing on extreme outcomes can produce different recommendations from rules that use all probability information. Decision B is favoured by a very optimistic view, decision C by a very cautious view and by regret minimisation, while decision A is favoured by the expected value criterion.

(C10) Evaluating all possibilities in the decision problem

Looking across all rules, the firm can evaluate each combined decision in terms of risk and reward. Decision A, premium revenue with stable costs, offers strong profits in favourable states without negative outcomes and achieves the highest expected profit, but it does not have the strongest worst-case profit. Decision B, premium revenue with flexible costs, has the highest possible profit when demand is high and costs are favourable but exposes

the firm to a negative profit of -200 when demand is weak and costs are unfavourable, and therefore has a high maximum regret. Decision C, volume revenue with stable costs, never performs extremely badly or extremely well. It offers the strongest worst-case profit and the smallest maximum regret, but its expected value is somewhat lower than that of decision A. Decision D, volume revenue with flexible costs, lies between the others in many states but does not dominate any of them in terms of either expected value, worst case or regret. This evaluation highlights how attitudes towards risk, regret and the role of probabilities lead to different recommended actions, and how separating revenue and expense decisions creates a richer set of trade-offs.

(C1) Formulating the final decision and summarising all criteria

Based on each of the decision criteria used in the solution, the recommended combined decision would be as follows. If the firm decides according to the Maximax rule and adopts a highly optimistic and risk-seeking attitude, it should choose decision B, premium revenue with flexible costs, because this combination can reach the highest profit among all combined states. If the firm decides according to the Maximin rule and wants to protect itself against the worst possible profit, it should choose decision C, volume revenue with stable costs, because this combination has the strongest worst-case outcome. If the firm uses the maximum opportunity or minimax regret criterion and cares mainly about limiting the largest possible regret from a wrong choice, it should again choose decision C, since its maximum regret is the smallest among all options. If the firm trusts the estimated probabilities for demand and cost conditions and uses the Bayes expected value rule to maximise expected annual profit, it should choose decision A, premium revenue with stable costs, because this combination has the highest expected monetary value.

The decision tree representation confirms how the two independent events, demand and costs, interact with the two separate decisions on revenue and expenses. The comparison and evaluation clarify how each decision rule reflects a particular attitude toward risk and uncertainty. Taking all criteria together, a coherent summary conclusion is that decision A, premium revenue with stable costs, is the most attractive option for a firm that wants to maximise expected profit while avoiding extreme downside risk, whereas decision C, volume revenue with stable costs, is more appropriate for a firm that prioritises robustness of the worst-case outcome and regret minimisation. For a firm that wishes to use probability information responsibly but still give weight to security, stating its final decision as choosing premium revenue with stable costs and clearly understanding how this differs from the more conservative choice volume revenue with stable costs is consistent with the overall analysis of this two-decision problem.

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2.4. Practice Activity 1

Problems and detailed worked solutions on Expected Value, Maximin, and Maximax.

2.4.1. Problem 1 — Local Café Investment

A small café is considering whether to introduce a new line of artisan desserts. The owner has two options:

- Launch the dessert line, in which case profits depend on demand:
 - High demand $\rightarrow$ 40 thousand USD
 - Low demand $\rightarrow$ 10 thousand USD
- Keep the current menu, yielding 25 thousand USD regardless of demand.

Market research indicates:

- Probability of high demand: 0.60
- Probability of low demand: 0.40

Question: Make a decision on the best option based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 1 — Local Café Investment

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Launch desserts; Keep current menu
States of nature	High demand (0.60); Low demand (0.40)
Events	Actual demand level observed after the decision
Consequences	Profit in thousands of USD for each alternative–state pair

(C3) Building the payoff table

Alternative	High demand (0.60)	Low demand (0.40)
Launch desserts	40	10
Keep menu	25	25

(C4) Applying the Maximax criterion The maximax rule ignores probabilities and selects the alternative with the highest possible payoff.

$$\max_i \text{Payoff}(\text{Launch}, S_i) = \max\{40, 10\} = 40$$

$$\max_i \text{Payoff}(\text{Keep}, S_i) = \max\{25, 25\} = 25$$

$$\max\{40, 25\} = 40$$

Because 40 is the largest best payoff, the maximax (risk-seeking) recommendation is launch desserts.

(C5) Applying the Maximin criterion The maximin rule focuses on the worst-case payoff of each alternative and picks the largest of those worst cases.

$$\min_i \text{Payoff}(\text{Launch}, S_i) = \min\{40, 10\} = 10$$

$$\min_i \text{Payoff}(\text{Keep}, S_i) = \min\{25, 25\} = 25$$

$$\max\{10, 25\} = 25$$

Since 25 exceeds 10, the maximin (risk-averse) choice is keep the current menu.

(C1) Formulating the decision problem for an optimal choice We frame the decision as choosing the menu strategy that maximizes the café's objective. Alternatives are Launch desserts or Keep menu; the uncertain states are high or low demand with probabilities 0.60 and 0.40. The optimal decision under each criterion uses the following calculations.

$$EV_{\text{Launch}} = 0,60(40) + 0,40(10)$$

$$= 0,60 \cdot 40 + 0,40 \cdot 10$$

$$= 24 + 4 = 28$$

$$EV_{\text{Keep}} = 0,60(25) + 0,40(25)$$

$$= 0,60 \cdot 25 + 0,40 \cdot 25$$

$$= 15 + 10 = 25$$

$$\max\{28, 25\} = 28$$

Summary of recommendations across decision criteria

- Maximax favors launch desserts (highest payoff of 40).
- Maximin favors keep the current menu (best worst-case payoff of 25).
- Expected value favors launch desserts (EV of 28).

Comparing the criteria shows that the launch option leads on optimistic and average performance, while keeping the menu provides stronger downside protection.

2.4.2. Problem 2 — Green Energy Production Mix

A renewable-energy company is evaluating its energy production strategy for the coming year. Three weather conditions are possible:

- Windy
- Sunny
- Cloudy

The company is comparing three strategies:

1. Build wind turbines
2. Build solar farms
3. Build a balanced hybrid system

Expected profits (in million USD):

- Wind turbines → 85 (Windy), 40 (Sunny), 25 (Cloudy)
- Solar farms → 30 (Windy), 90 (Sunny), 20 (Cloudy)
- Hybrid system → 60 (Windy), 60 (Sunny), 55 (Cloudy)

Weather probabilities:

- Windy → 0.30
- Sunny → 0.50

- Cloudy $\rightarrow 0.20$

Question: Make a decision on the best strategy based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 2 — Green Energy Production Mix

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Wind turbines; Solar farms; Hybrid system
States of nature	Windy (0.30); Sunny (0.50); Cloudy (0.20)
Events	Weather outcome driving power generation
Consequences	Profit in million USD for each strategy-weather combination

(C3) Building the payoff table

Strategy	Windy (0.30)	Sunny (0.50)	Cloudy (0.20)
Wind turbines	85	40	25
Solar farms	30	90	20
Hybrid system	60	60	55

(C4) Applying the Maximax criterion Maximax considers only the best possible outcome for each alternative and then picks the largest one.

$$\max_i \text{Payoff}(\text{Wind}, S_i) = \max\{85, 40, 25\} = 85$$

$$\max_i \text{Payoff}(\text{Solar}, S_i) = \max\{30, 90, 20\} = 90$$

$$\max_i \text{Payoff}(\text{Hybrid}, S_i) = \max\{60, 60, 55\} = 60$$

$$\max\{85, 90, 60\} = 90$$

The largest best payoff is 90, so the maximax (risk-seeking) decision is build solar farms.

(C5) Applying the Maximin criterion Maximin evaluates the worst payoff for each alternative and favors the highest of those worst cases.

$$\min_i \text{Payoff}(\text{Wind}, S_i) = \min\{85, 40, 25\} = 25$$

$$\min_i \text{Payoff}(\text{Solar}, S_i) = \min\{30, 90, 20\} = 20$$

$$\min_i \text{Payoff}(\text{Hybrid}, S_i) = \min\{60, 60, 55\} = 55$$

$$\max\{25, 20, 55\} = 55$$

Because 55 is the largest minimum payoff, the maximin (risk-averse) recommendation is the hybrid system.

(C1) Formulating the decision problem for an optimal choice The firm wants the production mix that best supports its profit goal. Alternatives are wind, solar, or a hybrid system and the uncertain states are windy, sunny, or cloudy with respective probabilities 0.30, 0.50, and 0.20. Optimizing by expected value yields:

$$\begin{aligned} EV_{\text{Wind}} &= 0,30(85) + 0,50(40) + 0,20(25) \\ &= 0,30 \cdot 85 + 0,50 \cdot 40 + 0,20 \cdot 25 \\ &= 25,5 + 20 + 5 = 50,5 \end{aligned}$$

$$\begin{aligned} EV_{\text{Solar}} &= 0,30(30) + 0,50(90) + 0,20(20) \\ &= 0,30 \cdot 30 + 0,50 \cdot 90 + 0,20 \cdot 20 \\ &= 9 + 45 + 4 = 58 \end{aligned}$$

$$\begin{aligned} EV_{\text{Hybrid}} &= 0,30(60) + 0,50(60) + 0,20(55) \\ &= 0,30 \cdot 60 + 0,50 \cdot 60 + 0,20 \cdot 55 \\ &= 18 + 30 + 11 = 59 \end{aligned}$$

$$\max\{50,5, 58, 59\} = 59$$

Summary of recommendations across decision criteria

- Maximax favors solar farms (highest payoff of 90).
- Maximin favors the hybrid system (best worst-case payoff of 55).
- Expected value favors the hybrid system (EV of 59).

The comparison shows solar excels only in the best-case outcome, while the hybrid strategy dominates on downside protection and average profitability.

2.4.3. Problem 3 — Global Shipping Network Design

A multinational logistics firm must redesign its global shipping network for the next five years to balance efficiency with resilience. Management is weighing three mutually exclusive configurations: a centralized network built around a single mega-hub in East Asia, a regional network with three hubs in the Americas, Europe, and Asia, and an adaptive AI-driven network that relies on dynamic routing powered by machine-learning models.

The firm is uncertain about future trade volatility. It models three possible scenarios: low volatility with stable trade flows, medium volatility with moderate regional disruptions, and high volatility with significant unpredictability. The firm assigns probabilities of 0.25 to low volatility, 0.45 to medium volatility, and 0.30 to high volatility. Expected profits (in million USD) are estimated at 210, 130, and -40 for the centralized option under low, medium, and high volatility; 180, 160, and 70 for the regional option; and 150, 175, and 140 for the AI-driven option.

Question: Make a decision on the best configuration based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 3 — Global Shipping Network Design

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Centralized hub; Regional hubs; AI-driven adaptive network
States of nature	Low volatility (0.25); Medium (0.45); High (0.30)
Events	Actual trade volatility realized over the planning horizon
Consequences	Profit in million USD for each configuration-volatility scenario

(C3) Building the payoff table

Configuration	Low volatility (0.25)	Medium volatility (0.45)	High volatility (0.30)
Centralized	210	130	-40
Regional	180	160	70
AI-Driven	150	175	140

(C4) Applying the Maximax criterion Maximax selects the alternative with the highest possible payoff regardless of downside.

$$\max_i \text{Payoff}(\text{Centralized}, S_i) = \max\{210, 130, -40\} = 210$$

$$\max_i \text{Payoff}(\text{Regional}, S_i) = \max\{180, 160, 70\} = 180$$

$$\max_i \text{Payoff}(\text{AI}, S_i) = \max\{150, 175, 140\} = 175$$

$$\max\{210, 180, 175\} = 210$$

The greatest best payoff is 210, so the maximax choice is the centralized network.

(C5) Applying the Maximin criterion Maximin emphasizes protection against the worst case by comparing the minimum payoff of each alternative.

$$\min_i \text{Payoff}(\text{Centralized}, S_i) = \min\{210, 130, -40\} = -40$$

$$\min_i \text{Payoff}(\text{Regional}, S_i) = \min\{180, 160, 70\} = 70$$

$$\min_i \text{Payoff}(\text{AI}, S_i) = \min\{150, 175, 140\} = 140$$

$$\max\{-40, 70, 140\} = 140$$

Because 140 is the largest of these minimums, the maximin recommendation is the AI-driven network.

(C1) Formulating the decision problem for an optimal choice The logistics firm frames its decision as selecting the network that maximizes long-run profit. Alternatives are centralized, regional, or AI-driven and the uncertain states are low, medium, or high volatility with probabilities 0.25, 0.45, and 0.30. Using expected value to target optimality:

$$\begin{aligned}
EV_{\text{Centralized}} &= 0,25(210) + 0,45(130) + 0,30(-40) \\
&= 0,25 \cdot 210 + 0,45 \cdot 130 + 0,30 \cdot (-40) \\
&= 52,5 + 58,5 - 12 = 99
\end{aligned}$$

$$\begin{aligned}
EV_{\text{Regional}} &= 0,25(180) + 0,45(160) + 0,30(70) \\
&= 0,25 \cdot 180 + 0,45 \cdot 160 + 0,30 \cdot 70 \\
&= 45 + 72 + 21 = 138
\end{aligned}$$

$$\begin{aligned}
EV_{\text{AI}} &= 0,25(150) + 0,45(175) + 0,30(140) \\
&= 0,25 \cdot 150 + 0,45 \cdot 175 + 0,30 \cdot 140 \\
&= 37,5 + 78,75 + 42 = 158,25
\end{aligned}$$

$$\text{máx}\{99, 138, 158,25\} = 158,25$$

Summary of recommendations across decision criteria

- Maximax favors the centralized network (highest payoff of 210).
- Maximin favors the AI-driven network (best worst-case payoff of 140).
- Expected value favors the AI-driven network (EV of 158.25).

The centralized design wins only on peak upside, while the AI-driven option provides the strongest performance on both downside risk and average returns across volatility scenarios.

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2.5. Practice Activity 2

Problems and worked solutions on decision rules with revenue and cost scenarios.

2.5.1. Problem 1 — Food Truck Menu Upgrade

A food truck is deciding whether to expand its menu or keep a classic menu. Annual demand and cost conditions are uncertain.

Two strategies are being compared:

- Expanded menu
- Classic menu

Revenue depends on demand (in thousand USD):

- High demand (0.50): Expanded 30, Classic 22
- Low demand (0.50): Expanded 15, Classic 18

Costs depend on input prices (in thousand USD):

- Favourable costs (0.60): Expanded 12, Classic 8
- Unfavourable costs (0.40): Expanded 18, Classic 10

Demand and cost conditions are independent.

Question: Make a decision on the best option based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 1 — Food Truck Menu Upgrade

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Expanded menu; Classic menu
States of nature	High/low demand combined with favourable/unfavourable costs
Events	Demand level and cost level realized after the decision
Consequences	Profit (revenue - costs) for each strategy and combined state
Probabilities	Demand: 0.50 high, 0.50 low; Costs: 0.60 favourable, 0.40 unfavourable

(C3) Building the payoff table from revenue and cost data Revenue by demand level (thousand USD):

Demand level	Probability	Expanded	Classic
High demand	0.50	30	22
Low demand	0.50	15	18

Costs by operating conditions (thousand USD):

Cost level	Probability	Expanded	Classic
Favourable costs	0.60	12	8
Unfavourable costs	0.40	18	10

Combined states of nature (independent demand and cost):

- S1: High demand + favourable costs, $P = 0.50 \times 0.60 = 0.30$
- S2: High demand + unfavourable costs, $P = 0.50 \times 0.40 = 0.20$
- S3: Low demand + favourable costs, $P = 0.50 \times 0.60 = 0.30$
- S4: Low demand + unfavourable costs, $P = 0.50 \times 0.40 = 0.20$

Preliminary payoff table (profit = revenue - costs):

State	Scenario	Expanded	Classic
S1	High demand, favourable costs	30 – 12	22 – 8
S2	High demand, unfavourable costs	30 – 18	22 – 10
S3	Low demand, favourable costs	15 – 12	18 – 8
S4	Low demand, unfavourable costs	15 – 18	18 – 10

Payoff table (profit = revenue - costs):

State	Scenario	Expanded	Classic
S1	High demand, favourable costs	18	14
S2	High demand, unfavourable costs	12	12
S3	Low demand, favourable costs	3	10
S4	Low demand, unfavourable costs	-3	8

(C4) Applying the Maximax criterion Maximax selects the alternative with the highest best payoff.

$$\max_i \text{Payoff}(\text{Expanded}, S_i) = \max\{18, 12, 3, -3\} = 18$$

$$\max_i \text{Payoff}(\text{Classic}, S_i) = \max\{14, 12, 10, 8\} = 14$$

$$\max\{18, 14\} = 18$$

The maximax (risk-seeking) decision is expand the menu.

(C5) Applying the Maximin criterion Maximin compares each alternative's worst payoff.

$$\min_i \text{Payoff}(\text{Expanded}, S_i) = \min\{18, 12, 3, -3\} = -3$$

$$\min_i \text{Payoff}(\text{Classic}, S_i) = \min\{14, 12, 10, 8\} = 8$$

$$\max\{-3, 8\} = 8$$

The maximin (risk-averse) decision is keep the classic menu.

(C1) Formulating the decision problem for an optimal choice The truck wants the menu that maximizes expected profit given the joint demand-cost probabilities.

$$\begin{aligned} EV_{\text{Expanded}} &= 0,30(18) + 0,20(12) + 0,30(3) + 0,20(-3) \\ &= 5,4 + 2,4 + 0,9 - 0,6 = 8,1 \end{aligned}$$

$$\begin{aligned} EV_{\text{Classic}} &= 0,30(14) + 0,20(12) + 0,30(10) + 0,20(8) \\ &= 4,2 + 2,4 + 3 + 1,6 = 11,2 \end{aligned}$$

$$\max\{8,1, 11,2\} = 11,2$$

Summary of recommendations across decision criteria

- Maximax favors expanded menu (best payoff of 18).
- Maximin favors classic menu (best worst-case payoff of 8).
- Expected value favors classic menu (EV of 11.2).

The expanded menu offers the highest upside, but the classic menu provides stronger protection and higher average profit.

2.5.2. Problem 2 — Subscription Bundle Choices

A streaming service is deciding between three subscription bundles for the next year. Revenue depends on demand, while costs depend on content-licensing prices.

Strategies:

1. Premium bundle
2. Standard bundle
3. Budget bundle

Revenue by demand level (in million USD):

- High demand (0.40): Premium 60, Standard 50, Budget 40
- Low demand (0.60): Premium 30, Standard 35, Budget 32

Costs by licensing conditions (in million USD):

- Favourable costs (0.50): Premium 25, Standard 20, Budget 18
- Unfavourable costs (0.50): Premium 35, Standard 28, Budget 24

Demand and cost conditions are independent.

Question: Make a decision on the best option based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 2 — Subscription Bundle Choices

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Premium; Standard; Budget bundles
States of nature	High/low demand combined with favourable/unfavourable licensing costs
Events	Demand outcome and licensing cost conditions for the year
Consequences	Profit (revenue - costs) under each bundle and combined state
Probabilities	Demand: 0.40 high, 0.60 low; Costs: 0.50 favourable, 0.50 unfavourable

(C3) Building the payoff table from revenue and cost data Revenue by demand level (million USD):

Demand level	Probability	Premium	Standard	Budget
High demand	0.40	60	50	40
Low demand	0.60	30	35	32

Costs by licensing conditions (million USD):

Cost level	Probability	Premium	Standard	Budget
Favourable costs	0.50	25	20	18
Unfavourable costs	0.50	35	28	24

Combined states of nature (independent demand and cost):

- S1: High demand + favourable costs, $P = 0.40 \times 0.50 = 0.20$
- S2: High demand + unfavourable costs, $P = 0.40 \times 0.50 = 0.20$
- S3: Low demand + favourable costs, $P = 0.60 \times 0.50 = 0.30$
- S4: Low demand + unfavourable costs, $P = 0.60 \times 0.50 = 0.30$

Preliminary payoff table (profit = revenue - costs):

State	Scenario	Premium	Standard	Budget
S1	High demand, favourable costs	60 – 25	50 – 20	40 – 18
S2	High demand, unfavourable costs	60 – 35	50 – 28	40 – 24
S3	Low demand, favourable costs	30 – 25	35 – 20	32 – 18
S4	Low demand, unfavourable costs	30 – 35	35 – 28	32 – 24

Payoff table (profit = revenue - costs):

State	Scenario	Premium	Standard	Budget
S1	High demand, favourable costs	35	30	22
S2	High demand, unfavourable costs	25	22	16
S3	Low demand, favourable costs	5	15	14
S4	Low demand, unfavourable costs	-5	7	8

(C4) Applying the Maximax criterion Maximax compares the best payoff for each bundle.

$$\max_i \text{Payoff}(\text{Premium}, S_i) = \max\{35, 25, 5, -5\} = 35$$

$$\max_i \text{Payoff}(\text{Standard}, S_i) = \max\{30, 22, 15, 7\} = 30$$

$$\max_i \text{Payoff}(\text{Budget}, S_i) = \max\{22, 16, 14, 8\} = 22$$

$$\max\{35, 30, 22\} = 35$$

The maximax (risk-seeking) choice is the premium bundle.

(C5) Applying the Maximin criterion Maximin selects the highest worst-case payoff.

$$\min_i \text{Payoff}(\text{Premium}, S_i) = \min\{35, 25, 5, -5\} = -5$$

$$\min_i \text{Payoff}(\text{Standard}, S_i) = \min\{30, 22, 15, 7\} = 7$$

$$\min_i \text{Payoff}(\text{Budget}, S_i) = \min\{22, 16, 14, 8\} = 8$$

$$\max\{-5, 7, 8\} = 8$$

The maximin (risk-averse) choice is the budget bundle.

(C1) Formulating the decision problem for an optimal choice The service wants the bundle with the highest expected profit under the joint probabilities.

$$\begin{aligned} EV_{\text{Premium}} &= 0,20(35) + 0,20(25) + 0,30(5) + 0,30(-5) \\ &= 7 + 5 + 1,5 - 1,5 = 12 \end{aligned}$$

$$\begin{aligned} EV_{\text{Standard}} &= 0,20(30) + 0,20(22) + 0,30(15) + 0,30(7) \\ &= 6 + 4,4 + 4,5 + 2,1 = 17 \end{aligned}$$

$$\begin{aligned} EV_{\text{Budget}} &= 0,20(22) + 0,20(16) + 0,30(14) + 0,30(8) \\ &= 4,4 + 3,2 + 4,2 + 2,4 = 14,2 \end{aligned}$$

$$\max\{12, 17, 14,2\} = 17$$

Summary of recommendations across decision criteria

- Maximax favors the premium bundle (best payoff of 35).
- Maximin favors the budget bundle (best worst-case payoff of 8).
- Expected value favors the standard bundle (EV of 17).

The standard bundle delivers the strongest average profit, even though premium wins on upside and budget protects the downside.

2.5.3. Problem 3 — Regional Distribution Plan

A logistics firm is choosing a distribution plan for the next three years and wants a strategy that can handle uncertain shipment demand and volatile operating costs. Management is considering three alternatives: building a single central hub, investing in multiple regional hubs, or relying on flexible partner networks.

The firm models demand as either high or low. If demand is high, the expected annual revenues (in million USD) are 120 for a central hub, 100 for regional hubs, and 90 for flexible partners, with a 0.30 probability. If demand is low, revenues are expected to be 70, 80, and 75 respectively, with a 0.70 probability. Independently, operating costs can be favourable or unfavourable. Under favourable costs (probability 0.40), annual costs are estimated at 50 for a central hub, 45 for regional hubs, and 40 for flexible partners. Under unfavourable costs (probability 0.60), annual costs are 80, 65, and 55 in the same order. The firm assumes demand conditions and cost conditions are independent.

Question: Make a decision on the best option based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution 3 — Regional Distribution Plan

Structured solution

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Central hub; Regional hubs; Flexible partners
States of nature	High/low demand combined with favourable/unfavourable operating costs
Events	Demand outcome and cost conditions over the planning period
Consequences	Profit (revenue - costs) for each plan and combined state
Probabilities	Demand: 0.30 high, 0.70 low; Costs: 0.40 favourable, 0.60 unfavourable

(C3) Building the payoff table from revenue and cost data Revenue by demand level (million USD):

Demand level	Probability	Central	Regional	Flexible
High demand	0.30	120	100	90
Low demand	0.70	70	80	75

Costs by operating conditions (million USD):

Cost level	Probability	Central	Regional	Flexible
Favourable costs	0.40	50	45	40
Unfavourable costs	0.60	80	65	55

Combined states of nature (independent demand and cost):

- S1: High demand + favourable costs, $P = 0.30 \times 0.40 = 0.12$
- S2: High demand + unfavourable costs, $P = 0.30 \times 0.60 = 0.18$
- S3: Low demand + favourable costs, $P = 0.70 \times 0.40 = 0.28$
- S4: Low demand + unfavourable costs, $P = 0.70 \times 0.60 = 0.42$

Preliminary payoff table (profit = revenue - costs):

State	Scenario	Central	Regional	Flexible
S1	High demand, favourable costs	120 – 50	100 – 45	90 – 40
S2	High demand, unfavourable costs	120 – 80	100 – 65	90 – 55
S3	Low demand, favourable costs	70 – 50	80 – 45	75 – 40
S4	Low demand, unfavourable costs	70 – 80	80 – 65	75 – 55

Payoff table (profit = revenue - costs):

State	Scenario	Central	Regional	Flexible
S1	High demand, favourable costs	70	55	50
S2	High demand, unfavourable costs	40	35	35
S3	Low demand, favourable costs	20	35	35
S4	Low demand, unfavourable costs	-10	15	20

(C4) Applying the Maximax criterion Maximax compares the best profit for each plan.

$$\max_i \text{Payoff}(\text{Central}, S_i) = \max\{70, 40, 20, -10\} = 70$$

$$\max_i \text{Payoff}(\text{Regional}, S_i) = \max\{55, 35, 35, 15\} = 55$$

$$\max_i \text{Payoff}(\text{Flexible}, S_i) = \max\{50, 35, 35, 20\} = 50$$

$$\max\{70, 55, 50\} = 70$$

The maximax (risk-seeking) choice is the central hub plan.

(C5) Applying the Maximin criterion Maximin picks the highest of the worst-case profits.

$$\min_i \text{Payoff}(\text{Central}, S_i) = \min\{70, 40, 20, -10\} = -10$$

$$\min_i \text{Payoff}(\text{Regional}, S_i) = \min\{55, 35, 35, 15\} = 15$$

$$\min_i \text{Payoff}(\text{Flexible}, S_i) = \min\{50, 35, 35, 20\} = 20$$

$$\max\{-10, 15, 20\} = 20$$

The maximin (risk-averse) choice is the flexible partners plan.

(C1) Formulating the decision problem for an optimal choice The firm compares expected profits using joint probabilities for each demand-cost state.

$$EV_{\text{Central}} = 0,12(70) + 0,18(40) + 0,28(20) + 0,42(-10)$$

$$= 8,4 + 7,2 + 5,6 - 4,2 = 17$$

$$EV_{\text{Regional}} = 0,12(55) + 0,18(35) + 0,28(35) + 0,42(15)$$

$$= 6,6 + 6,3 + 9,8 + 6,3 = 29$$

$$EV_{\text{Flexible}} = 0,12(50) + 0,18(35) + 0,28(35) + 0,42(20)$$

$$= 6 + 6,3 + 9,8 + 8,4 = 30,5$$

$$\text{máx}\{17, 29, 30,5\} = 30,5$$

Summary of recommendations across decision criteria

- Maximax favors the central hub plan (best payoff of 70).
- Maximin favors the flexible partners plan (best worst-case payoff of 20).
- Expected value favors the flexible partners plan (EV of 30.5).

The flexible plan gives the best average outcome while maintaining strong downside protection against low demand and high costs.

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Capítulo 3

Grade 10 - 3rd Term

3.1. Grade 10: Continuous Probability Distributions

Complex and non-standard problems solving.

Evaluate finance context situations with normal distribution and other continuous random variables.

- Area of simple geometric figures.
- Random variable.
- Continuous probability distribution.
- Mean and variance.
- Normal distribution.
- Exponential distribution.
- t-student distribution.
- Investment portfolios.
- Risk.

3.1.1. Assessment Criteria

1. Describes probabilities in continuous variables as a density function.
2. Explains how a uniform random variable works.
3. Expresses how to work with distributions that have a linear behaviour.

4. Interprets the concept of variance and standard deviation of a probability function.
5. Uses the normal standard distribution and its tables to calculate probabilities in context situations.
6. Employs the standardisation of the normal variable in real life problems.
7. Solves problems through the concept of inverse normal distribution.
8. Summarises the elements and characteristics that an investment portfolio may have.
9. Establishes that the risk of a portfolio is determined by its probability distribution.
10. Evaluates finance context situations with normal distribution and other continuous random variables.

3.1.2. Support Materials

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Capítulo 4

Grade 11 - 1st Term

4.1. Grade 11: Continuous Probability Distributions

Complex and non-standard problems solving.

Evaluate finance context situations with normal distribution and other continuous random variables.

- Area of simple geometric figures.
- Random variable.
- Continuous probability distribution.
- Mean and variance.
- Normal distribution.
- Exponential distribution.
- t-student distribution.
- Investment portfolios.
- Risk.

4.1.1. Assessment Criteria

1. Describes continuous random variables considering the concept.
2. Interprets probabilities in continuous variables as the area under the curve.
3. Explains how a uniform random variable works.

4. Shows how to work with distributions that have other shapes, like triangular or circular.
5. Uses the normal distribution and its tables to calculate probabilities in context situations.
6. Employs the standardization of the normal variable in complex problems.
7. Solves problems through the concept of inverse normal distribution.
8. Differentiates the elements and characteristics that an investment portfolio may have.
9. Establishes the risk of a portfolio based on its probability distribution.
10. Evaluates finance context situations with normal distribution and other continuous random variables.

4.1.2. Support Materials

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Capítulo 5

Grade 11 - 2nd Term

5.1. Grade 11: Confidence & Interval Estimation

Inferential tools for samples and populations using normal and t-student approaches in Global Economics.

Confidence intervals translate sample information into ranges of plausible population values, supporting economic and financial planning under uncertainty.

- Area of simple geometric figures.
- Sample and population.
- Random variable.
- Continuous probability distribution.
- Mean and variance.
- Normal distribution.
- t-student distribution.
- Investment portfolios.
- Point vs. interval: a single-value estimate is precise but uncertain; an interval brackets plausible values with a stated confidence level.
- Known variance: use Z-based critical values when population variance is available.
- Unknown variance: use t-student with sample standard deviation and degrees of freedom.

5.1.1. Assessment Criteria

1. Recognises the concept of mean and standard deviation in a sample as well as in a population.
2. Distinguishes between statistic of a population and estimating the statistic.
3. Explains the difference between point estimation vs. interval estimation for a statistic in a population.
4. Interprets the concept of an interval of confidence in real life applications of economy and finance.
5. Calculates the confidence interval for the mean when the variance is known.
6. Establishes the sample size so the error will not exceed a specified amount.
7. Constructs the confidence interval for a proportion.
8. Calculates confidence intervals applying t-student distribution in context situations.
9. Estimates the confidence interval for the mean when the variance is unknown in context situations.
10. Concludes about a statistical parameter in situations in finance and economy.

5.1.2. Support Materials

- [Distributions \(Web Page\)](#): reference page with interactive probability distributions.
- [Interactive Confidence Interval Tool \(Web Page\)](#): tool to upload data and compute confidence intervals visually.
- [Practice Sets \(Web Page\)](#): extra practice to apply confidence interval skills and interpretation.

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5.2. Formulas for Confidence & Interval Estimation

5.2.1. Getting Oriented: Random Variables and Notation

Suppose the same process is repeated and gives similar kinds of results. For example, two small datasets from the same distribution could look like:

- Dataset A (5 quiz scores): 6, 7, 8, 7, 6.
- Dataset B (5 quiz scores): 7, 6, 8, 7, 7.

Each score is one outcome of the same random process. We describe that process with a random variable. For example, let X be the quiz score of a randomly chosen student. Then the values we actually see are written as x or x_i , such as 6, 7, or 8.

Standard notation used in the formulas:

- X (uppercase): the random variable itself.
- x or x_i (lowercase): a specific observed value.
- $X_1, X_2, \dots, X_n$: outcomes from repeated observations of the same process.
- $x_1, x_2, \dots, x_n$: the actual numbers recorded in a dataset.

Before solving the examples, we work with a generic dataset made of n data points: $x_1, x_2, \dots, x_n$, where x_i is the value of the i -th data point in the sample (for example, the score of the i -th student in a quiz, or the i -th measurement taken in an experiment). All the formulas below refer to this dataset.

Use $X_1, X_2, \dots$ to label the first, second, and subsequent observations by position in the dataset, and $x_1, x_2, \dots$ for the numerical values those observations take. The indices are fixed by position, while the values can change from sample to sample. For example, in one sample the first observation might be $x_1^{(1)} = 1$, while in another sample the first observation could be $x_1^{(2)} = 5$.

5.2.2. Sample Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Meaning of variables:

- x_i : each observed data value.
- n : number of observations in the sample.
- $\bar{x}$: the sample mean.

5.2.3. Standard Deviation

There are two closely related ways to compute the standard deviation, depending on whether the data represent a sample or the entire population:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Meaning of variables:

- x_i : each observed data value.
- $\bar{x}$: the sample mean.
- n : number of observations in the sample.
- N : number of observations in the full population.
- $n - 1$: degrees of freedom when using the data as a sample of a larger population.
- μ : population mean (used when the full population is known).
- s : sample standard deviation, used when the dataset is a sample and we estimate population variability (denominator $n - 1$).
- s^2 : sample variance, used when the dataset is a sample and we estimate population variability; it measures the average squared deviation of the observations from the sample mean (denominator $n - 1$).
- σ : population standard deviation, used when the dataset is the entire population (denominator N).

In summary, use s when your data are a sample from a larger population because the $n - 1$ denominator accounts for estimation uncertainty. Use σ when you have the complete population, where the denominator is N and no estimation is needed.

5.2.4. Bernoulli Distribution

Let X follow a Bernoulli distribution with success probability p . The average (mean) outcome is $\mu = \mathbb{E}[X] = p$, and the variance is:

$$s^2 = \text{Var}(X) = p(1 - p)$$

In words, a Bernoulli variable only takes values 0 or 1, so the average is the probability of success, and the variability depends on how close p is to 0 or 1.

From a sample of size n with x successes, estimate the probability of success with $\hat{p} = \frac{x}{n}$, which is the sample mean of the 0/1 outcomes.

5.2.5. Margin of Error for a Mean

$$E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Normal distribution critical values

Confidence level	z critical value
99 %	2.576
98 %	2.326
95 %	1.960
90 %	1.645

$$E = t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

Student's t distribution critical values

df	99 %	98 %	95 %	90 %
1	63.657	31.821	12.706	6.314
2	9.925	6.965	4.303	2.920
3	5.841	4.541	3.182	2.353
4	4.604	3.747	2.776	2.132
5	4.032	3.365	2.571	2.015
6	3.707	3.143	2.447	1.943
7	3.499	2.998	2.365	1.895
8	3.355	2.896	2.306	1.860
9	3.250	2.821	2.262	1.833
10	3.169	2.764	2.228	1.812
11	3.106	2.718	2.201	1.796
12	3.055	2.681	2.179	1.782
13	3.012	2.650	2.160	1.771
14	2.977	2.624	2.145	1.761
15	2.947	2.602	2.131	1.753
16	2.921	2.583	2.120	1.746
17	2.898	2.567	2.110	1.740
18	2.878	2.552	2.101	1.734
19	2.861	2.539	2.093	1.729
20	2.845	2.528	2.086	1.725
21	2.831	2.518	2.080	1.721
22	2.819	2.508	2.074	1.717
23	2.807	2.500	2.069	1.714
24	2.797	2.492	2.064	1.711
25	2.787	2.485	2.060	1.708
26	2.779	2.479	2.056	1.706
27	2.771	2.473	2.052	1.703
28	2.763	2.467	2.048	1.701
29	2.756	2.462	2.045	1.699
30	2.750	2.457	2.042	1.697

The correct distribution for the margin of error of a mean is the Student's t distribution with $n - 1$ degrees of freedom. As the sample size increases, the t distribution converges to the normal distribution, and for $n \geq 30$ it is nearly indistinguishable from the normal distribution for practical purposes.

Meaning of variables:

- $z_{\alpha/2}$: critical value from the standard normal distribution (for example, 1.96 for 95 % confidence).
- $t_{\alpha/2, n-1}$: critical value from the t distribution with $n - 1$ degrees of freedom.
- s : standard deviation computed from the data (typically using $n - 1$ in the denominator when treating the data as a sample).
- n : sample size.
- E : margin of error.

5.2.6. Confidence Interval for a Mean

$$\bar{x} \pm E$$

Meaning: the interval of plausible values for the population mean based on the sample mean and the margin of error.

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5.3. Main Examples

5.3.1. Example 1

Problem description

The school is planning next semester's scholarship fund and wants to estimate the average monthly scholarship cost per student. Teacher A oversees the STEM cohort, and Teacher B oversees the Arts cohort. Each teacher serves a different student population, so the finance office takes separate samples to estimate the average cost for each group. A confidence interval is needed because the fund must cover typical costs without overspending.

Variable measured: monthly scholarship cost per student (USD).

Planning decision: estimate how much money to allocate per student for each program so the total scholarship fund is realistic and conservative.

Sample data collected this semester:

- Teacher A (STEM cohort, sample size $n_A = 5$): 480, 520, 500, 510, 490.

- Teacher B (Arts cohort, sample size $n_B = 10$): 500, 520, 540, 510, 530, 550, 520, 510, 540, 530.

After the samples are collected, the principal shares the true population parameters for each cohort: $\mu_A = 505$, $\sigma_A = 20$, $\mu_B = 530$, and $\sigma_B = 18$. This makes it possible to compute results two ways: using the sample standard deviations with the t distribution (variance unknown) and using the true population standard deviations with the normal distribution (variance known).

Because each sample represents only part of its population, we use confidence intervals to express uncertainty and plan the scholarship fund with a safety margin.

(C1) Obtain sample mean and standard deviation

Each teacher summarises their sample with a mean and standard deviation to estimate typical monthly scholarship costs and the spread around those costs.

Teacher A (STEM cohort):

$$\begin{aligned}\bar{x}_A &= \frac{480 + 520 + 500 + 510 + 490}{5} = \frac{2500}{5} = 500 \\ s_A^2 &= \frac{(480 - 500)^2 + (520 - 500)^2 + (500 - 500)^2 + (510 - 500)^2 + (490 - 500)^2}{5 - 1} = \frac{1000}{4} = 250 \\ s_A &= \sqrt{250} \approx 15,81\end{aligned}$$

Teacher B (Arts cohort):

$$\begin{aligned}\bar{x}_B &= \frac{500 + 520 + 540 + 510 + 530 + 550 + 520 + 510 + 540 + 530}{10} = \frac{5250}{10} = 525 \\ s_B^2 &= \frac{\sum (x_i - 525)^2}{10 - 1} = \frac{2250}{9} = 250 \\ s_B &= \sqrt{250} \approx 15,81\end{aligned}$$

These sample statistics provide point estimates for the population means and standard deviations, which will be refined with confidence intervals.

(C2) Compare sample estimates across cohorts

Comparing the samples already suggests differences: Teacher B's sample mean is higher ($\bar{x}_B = 525$) than Teacher A's ($\bar{x}_A = 500$). Both samples show the same estimated variability ($s_A \approx s_B \approx 15,81$), but the smaller STEM sample will still have more sampling uncertainty. The principal's true parameters confirm the cohorts differ ($\mu_A = 505$ vs. $\mu_B = 530$), giving a benchmark to compare the t-based estimates to the variance-known results.

(C3) Explain the need for estimation based on the comparison

Because the school cannot survey every student in each cohort, the sample means and standard deviations are only estimates. Relying on a single number could lead to underfunding or overfunding the scholarship pool. Confidence intervals turn the point estimates into ranges that show plausible values for the true average monthly cost in each program, which is essential for responsible budgeting. The principal's shared μ and σ let the teachers verify the t-based intervals against variance-known Z intervals.

(C5) Calculate the confidence interval (variance known)

With the population variance known, use the Z distribution for a 95 % confidence level. For $\alpha = 0,05$, $z_{\alpha/2} = z_{0,025} = 1,96$. Using the population standard deviations:

$$E_A^{(Z)} = z_{\alpha/2} \cdot \frac{\sigma_A}{\sqrt{n_A}} = 1,96 \cdot \frac{20}{\sqrt{5}} \approx 17,53$$

$$\mu_A \in 500 \pm 17,53 = (482,47, 517,53)$$

$$E_B^{(Z)} = z_{\alpha/2} \cdot \frac{\sigma_B}{\sqrt{n_B}} = 1,96 \cdot \frac{18}{\sqrt{10}} \approx 11,16$$

$$\mu_B \in 525 \pm 11,16 = (513,84, 536,16)$$

(C8) Calculate the confidence interval (variance unknown)

Because the population variance is unknown at the time of sampling, use the t distribution with a 95 % confidence level for each cohort:

Teacher A (df = 4):

$$E_A = t_{0,025,4} \cdot \frac{s_A}{\sqrt{n_A}} \approx 2,776 \cdot \frac{15,81}{\sqrt{5}} \approx 2,776 \cdot 7,07 \approx 19,62$$

$$\mu_A \in 500 \pm 19,62 = (480,38, 519,62)$$

Teacher B (df = 9):

$$E_B = t_{0,025,9} \cdot \frac{s_B}{\sqrt{n_B}} \approx 2,262 \cdot \frac{15,81}{\sqrt{10}} \approx 2,262 \cdot 5,00 \approx 11,31$$

$$\mu_B \in 525 \pm 11,31 = (513,69, 536,31)$$

The larger sample for Teacher B yields a smaller margin of error and a narrower interval, reflecting greater precision.

(C4) Interpret and compare intervals for scholarship planning

The principal's true parameters allow a variance-known analysis using the Z distribution to compare with the earlier t-based estimates.

Variance unknown (t, 95 %) recap: Teacher A used $\mu_A \in (480,38, 519,62)$ and Teacher B used $\mu_B \in (513,69, 536,31)$ based on their sample standard deviations.

Variance known (Z, 95 %) recap: Using the population standard deviations, the Z-based intervals are slightly narrower, reinforcing the same ordering between cohorts.

For scholarship planning, the finance team can treat the upper ends of the intervals as conservative per-student allocations to avoid shortfalls. Teacher A's intervals are wider because the STEM sample is smaller, so the budget for that cohort needs a larger cushion. Teacher B's intervals are narrower thanks to the larger sample, giving planners more confidence in the Arts allocation. The comparison shows how larger samples reduce uncertainty and support more precise budgeting, while smaller samples imply more cautious funding decisions.

(C10) Conclude about a statistical parameter in finance and economy

Based on the confidence intervals, the finance office can conclude that the true mean monthly scholarship cost per student differs by cohort, with the STEM cohort centered near 500 USD and the Arts cohort centered near 525 USD. This supports budgeting separate per-student allocations for each program rather than assuming a single overall mean, reducing the risk of underfunding or overspending in the scholarship fund.

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5.3.2. Example 2

Problem description

A café wants to plan staffing and inventory for online orders. To estimate typical demand, the manager randomly samples 50 days of order counts and summarises them in grouped form below. The goal is to estimate the mean daily orders and build a 95 % confidence interval for decision making.

Known population values: true mean ($\mu = 11,4$), true standard deviation ($\sigma = 4,2$) (used here to check the sample-based estimates).

Sample taken now: the 50 days of data are recorded in grouped form:

Order count (x)	Frequency (f)
5	10
9	15
12	18
20	7

The owner wants to estimate the population mean number of daily orders and produce a 95 % confidence interval to guide staffing and stock decisions.

(C1) Obtain sample mean and standard deviation

First summarise the sample with a grouped mean to represent the population mean:

$$\sum fx = 10(5) + 15(9) + 18(12) + 7(20) = 50 + 135 + 216 + 140 = 541$$

$$\bar{x} = \frac{\sum fx}{n} = \frac{541}{50} = 10,82$$

Then measure variability using the grouped standard deviation to mirror the population spread:

$$\begin{aligned} \sum f(x - \bar{x})^2 &= 10(5 - 10,82)^2 + 15(9 - 10,82)^2 + 18(12 - 10,82)^2 + 7(20 - 10,82)^2 \\ &= 10(33,87) + 15(3,31) + 18(1,39) + 7(84,27) = 338,70 + 49,65 + 25,02 + 589,89 = 1003,38 \\ s &= \sqrt{\frac{\sum f(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{1003,38}{49}} \approx \sqrt{20,48} \approx 4,53 \end{aligned}$$

(C2) Compare sample values to the known true values

The grouped $\bar{x} = 10,82$ underestimates the true mean ($\mu = 11,4$) by about 0.6 orders, and $s \approx 4,53$ sits slightly above the true spread ($\sigma = 4,2$). This illustrates how a single sample can miss the exact population values.

(C3) Explain the need for estimation, based on the comparison

Because the café can only sample a subset of days, the sample has to stand in for the ongoing population. The gaps between the sample statistics and the known population values show why we rely on estimation: samples give approximate, not exact, answers.

(C5) Calculate the confidence interval (variance known)

Compute the confidence interval for the mean at the same 95 % confidence level using the known population variance.

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1,96 \cdot \frac{4,2}{\sqrt{50}} \approx 1,16$$

$$\mu \in 10,82 \pm 1,16 = [9,66, 11,98]$$

(C8) Calculate the confidence interval (variance unknown)

Compute the confidence interval for the mean at the same 95 % confidence level using the sample variance.

$$E = t_{\alpha/2,49} \cdot \frac{s}{\sqrt{n}} = t_{0,025,49} \cdot \frac{4,53}{\sqrt{50}} \approx 2,01 \cdot 0,64 \approx 1,29$$

$$\mu \in 10,82 \pm 1,29 = [9,53, 12,11]$$

The t-based interval is slightly wider because it relies on the sample standard deviation.

(C4) Interpret the confidence interval in real-life economic or financial terms

For staffing and inventory budgets, the café can plan using the lower limit (about 9.5–9.7 orders) to avoid overstocking while recognising that true demand could reach the upper limit (about 12.0–12.1). The interval turns statistical uncertainty into a practical spending range.

(C10) Conclude about a statistical parameter in finance and economy

The café can conclude that the true mean daily online order count is around 10–12 orders, with a reasonable planning value near 11 orders per day. This parameter estimate informs staffing levels and inventory targets so budgets match typical demand rather than extreme days.

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5.3.3. Example 3

Problem description

A regional economic survey asks 200 small-business owners whether they expect profits to grow next quarter. The goal is to estimate the population proportion who are optimistic about growth.

Sample results: 124 owners answered “yes.”

Construct a 95 % confidence interval for the population proportion of optimistic owners.

(C7) Compute the sample proportion, standard error, and confidence interval

The binary variable is whether an owner expects profits to grow (yes = success, no = failure). The sample proportion is

$$\hat{p} = \frac{124}{200} = 0,62$$

Because proportions come from binary outcomes, variability depends on $\hat{p}$ itself:

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0,62(0,38)}{200}} \approx 0,034$$

This shows that larger samples shrink the standard error and yield tighter intervals. For 95 % confidence, use $z_{\alpha/2} = 1,96$:

$$E = 1,96 \cdot 0,034 \approx 0,067$$

$$\hat{p} \pm E = 0,62 \pm 0,067 = (0,553, 0,687)$$

The interval gives the plausible range of the true population proportion.

(C4) Interpret the interval in economic or financial terms

With 95 % confidence, the true share of optimistic businesses lies between about 55.3 % and 68.7 %. Policy planners can use this range to estimate how many firms may seek expansion loans or increase hiring, aligning support programs with realistic demand.

(C10) Conclude about a statistical parameter in finance and economy

The survey supports concluding that the population proportion of optimistic small-business owners is roughly 0.55–0.69, with a central estimate around 0.62. Economic planners can use this parameter to forecast loan demand and design support programs that match expected business sentiment.

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5.3.4. Example 4

Problem description

A finance team is estimating the average monthly return (%) of a diversified investment portfolio based on a very large sample of return observations. Because the dataset is large, the t-student distribution can be approximated by the normal distribution, even though the population variance is unknown. The team wants to build a 95 % confidence interval for the mean return to guide risk and allocation decisions.

Given sample summary:

- Sample size ($n = 400$).
- Sample mean ($\bar{x} = 1,2 \%$).
- Sample standard deviation ($s = 2,5 \%$).

Estimate the 95 % confidence interval for the population mean monthly return (μ), and explain why the normal approximation is justified.

(C9) Estimate the mean with a large-sample t approximation

Because the population variance is unknown, the exact approach would use the t distribution with $n - 1$ degrees of freedom. However, since $n = 400$ is very large, the t distribution is almost identical to the normal distribution. Therefore we approximate using $z_{0,025} = 1,96$.

Compute the margin of error:

$$E \approx z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 1,96 \cdot \frac{2,5}{\sqrt{400}} = 1,96 \cdot \frac{2,5}{20} = 1,96 \cdot 0,125 = 0,245$$

So the confidence interval is:

$$\mu \in 1,2 \pm 0,245 = (0,955, 1,445)$$

This interval estimates the plausible range for the true mean monthly return.

(C4) Interpret the confidence interval for a finance decision

The interval suggests the portfolio's true mean monthly return is likely between about 0.96 % and 1.45 %. For allocation decisions, the finance team can use the lower bound as a conservative expected return when stress-testing plans and use the midpoint as a baseline forecast. The range also helps compare this portfolio to alternatives with different risk profiles.

(C10) Conclude about a statistical parameter in finance and economy

Based on the large sample, the team can conclude that the population mean monthly return is about 1.2 %, with a 95 % confidence interval from roughly 0.96 % to 1.45 %. Even though the variance is unknown, the large-sample approximation supports using the normal critical value to obtain a reliable interval for planning.

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Capítulo 6

Grade 11 - 3rd Term

6.1. Grade 11: Hypothesis Tests

Complex and non-standard problems solving.

Model a research question in which a statistical hypothesis must be tested.

- Area of simple geometric figures.
- Sample and population.
- Random variable.
- Continuous probability distribution.
- Mean and variance.
- Normal distribution.
- t-student distribution.
- Investment portfolios.
- Optimization problems with maxima and minima.

6.1.1. Assessment Criteria

1. Describes the concept of statistical hypothesis.
2. Identifies the null and alternative hypothesis in a statistical test.
3. Determines whether a test should be one tailed or two tailed.

4. Applies the general procedure for testing a hypothesis with fixed probability in context situations.
5. Calculates the hypothesis tests on the mean of a population.
6. Establishes the right hypothesis on a test about the proportion of a population.
7. Contrasts the P-values in hypotheses tests with the reference probability.
8. Analyses the relationship between hypothesis testing and confidence interval estimation.
9. Evaluates context situations in finance and economy in which a statistical hypothesis appears.
10. Models a research question in which a statistical hypothesis must be tested.

6.1.2. Support Materials

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